

General Circulation of the Atmosphere



Opening Comments

- ☞ The global system of atmospheric motions that is generated by the uneven heating of Earth's surface by the Sun is called the general circulation.
- ☞ Wind circulates in each hemisphere in three distinct cells which help transport energy and heat from the equator to the poles.
- ☞ The General Circulation of the Atmosphere refers to as the mean (time and zonal mean) condition of the three dimensional atmosphere structure over a long enough period.
- ☞ The major factors influences on the general circulation of the atmosphere are:
 - Differential heating
 - Rotation of the Earth
 - Topography
 - Atmospheric and oceanic fluid dynamics
- ☞ Theory of general circulation of the atmosphere is helpful to build the models of the Earth's atmosphere, both past, future

Some Nomenclature

$x \longrightarrow +ve$ eastward

$y \longrightarrow +ve$ northward

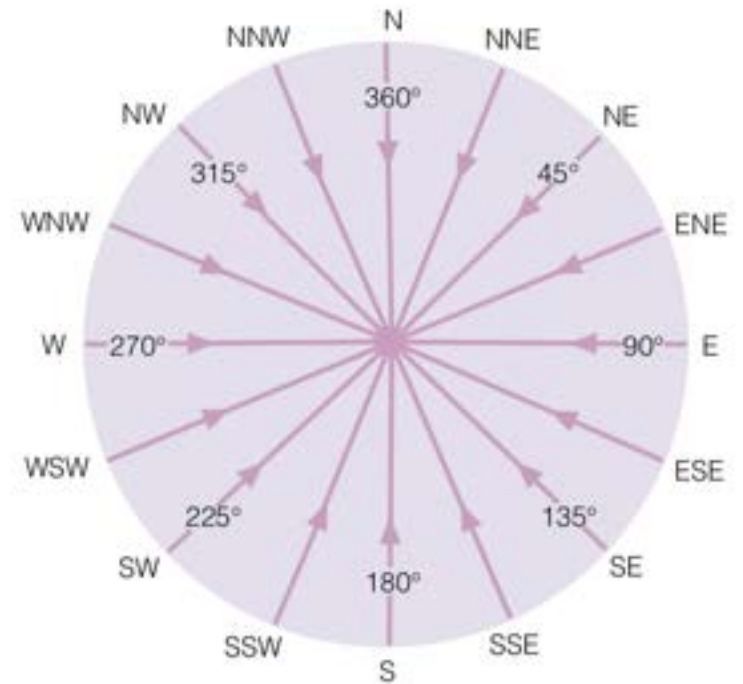
$z \longrightarrow +ve$ upward

Winds:

x - component : $u \longrightarrow$ often called **Zonal Winds**

y - component : $v \longrightarrow$ often called **Meridional Winds**

z - component : $w \longrightarrow$ **Vertical Velocity**



$u \longrightarrow +ve$: west to east winds are also called **WESTERLY**

$u \longrightarrow -ve$: east to west wind are also called **EASTERLY**

$v \longrightarrow +ve$ northward $\longrightarrow$ **SOUTHERLY**

$v \longrightarrow -ve$ southward $\longrightarrow$ **NORTHERLY**

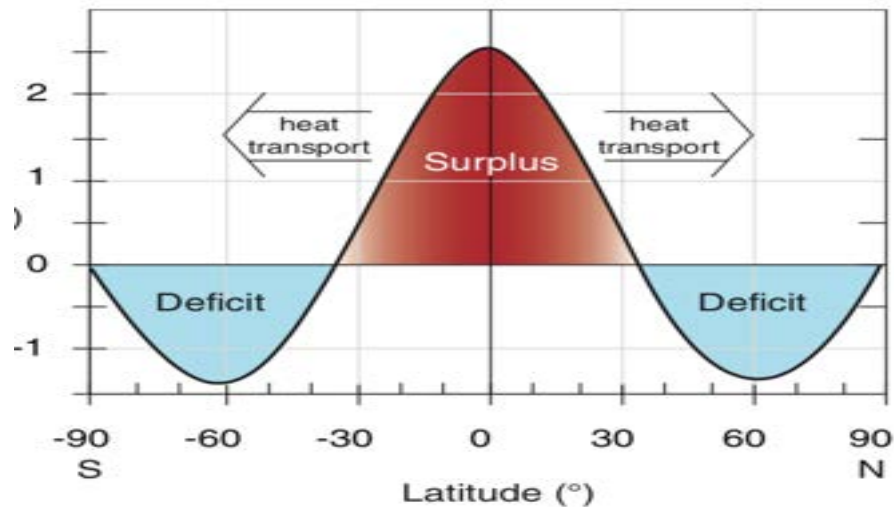
☞ Wind direction can be expressed in degrees

Variations in Heat Balance

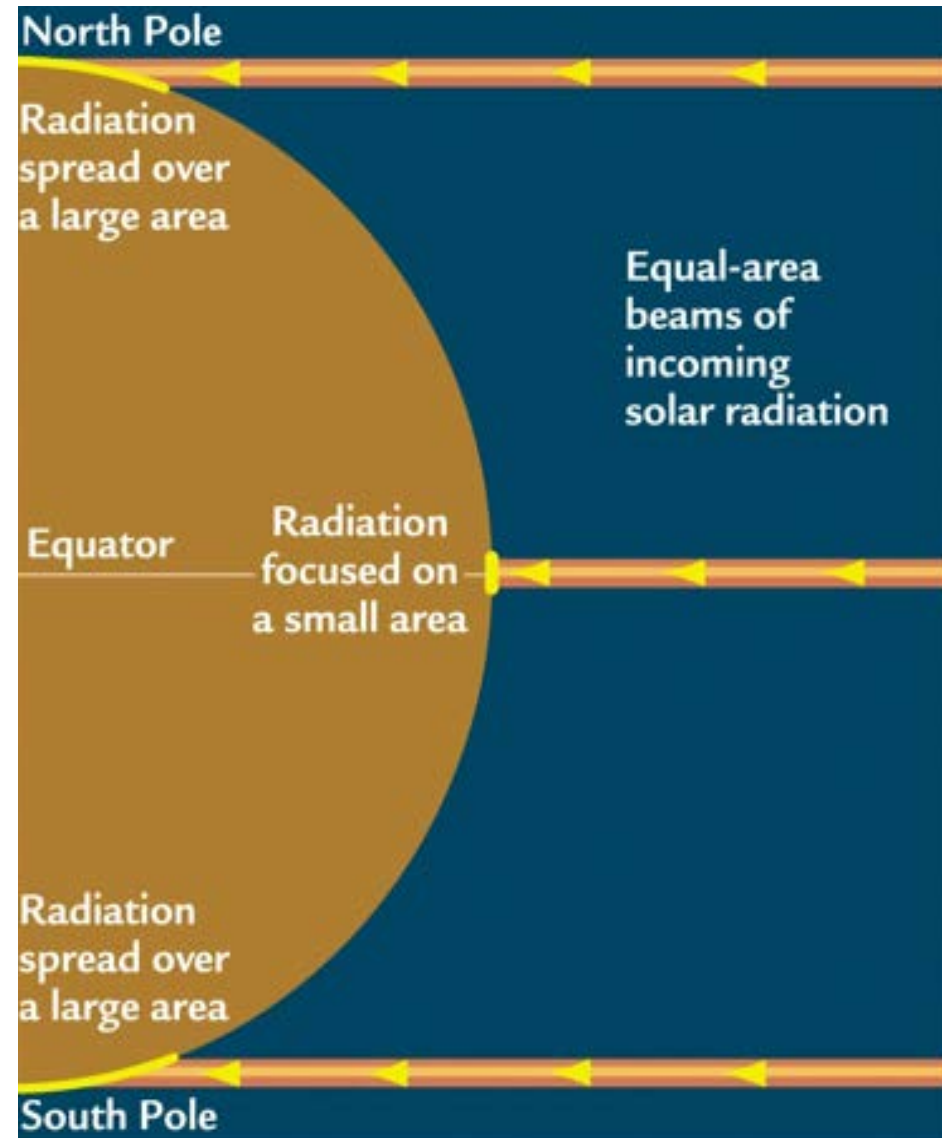
☞ Incoming solar radiation

- Stronger at low latitudes
- Weaker at high latitudes

☞ Tropics receive more solar radiation per unit area than Poles.



☞ Global Atmospheric Circulations are driven by the uneven horizontal distribution of the net incoming radiation.

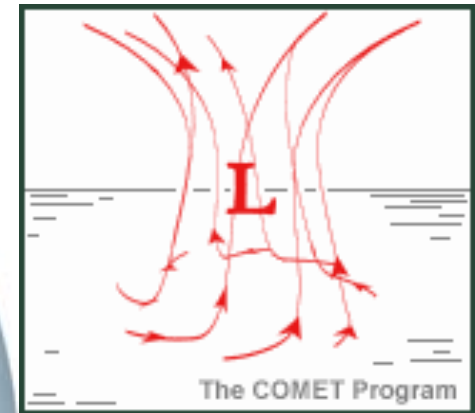
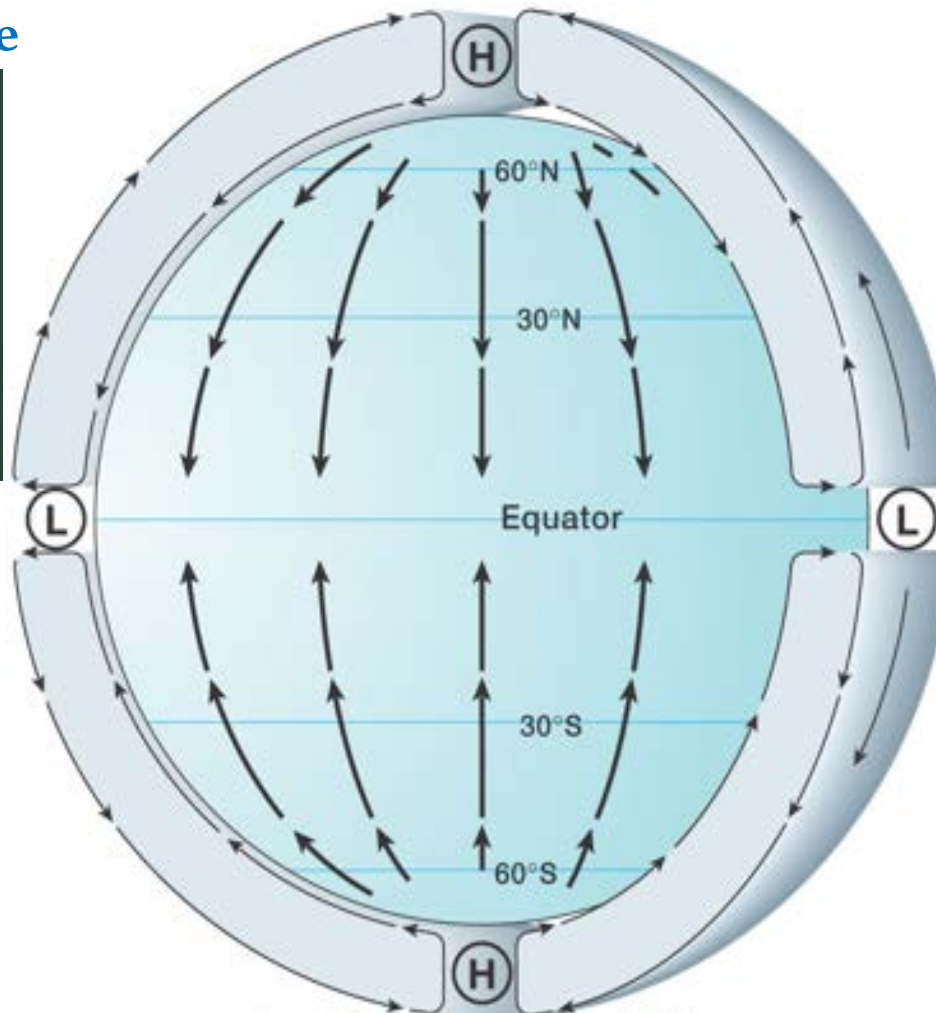
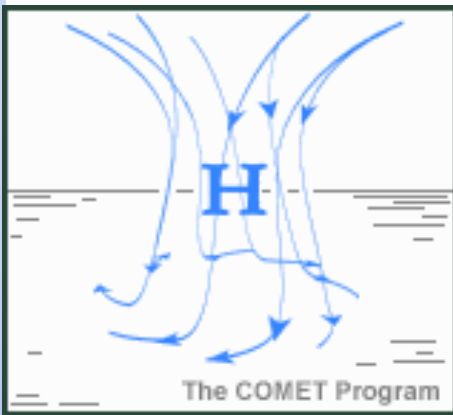


General Circulation-Hadley Cell (1735)



Basically, General circulation structure started with observations collected from the sailors (surface wind and pressure).

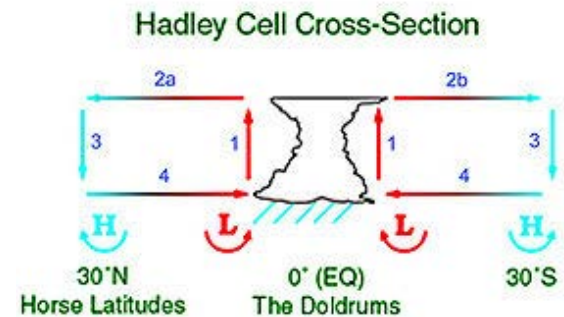
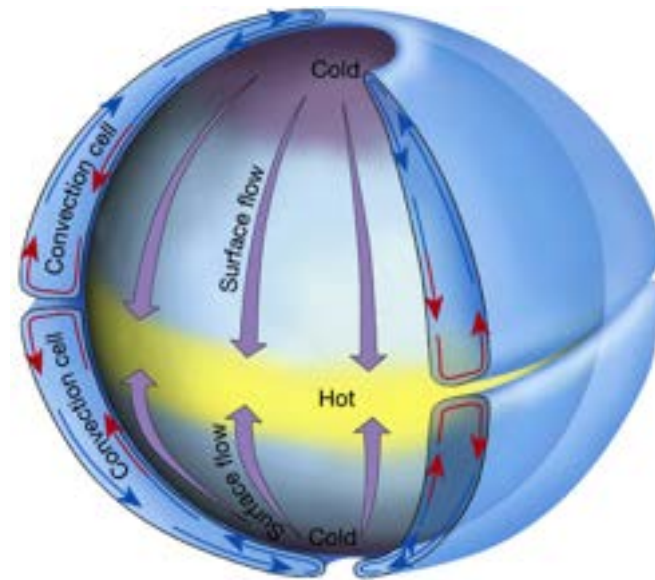
Sinking air associated with **High pressure**



Rising air associated with **Low pressure**

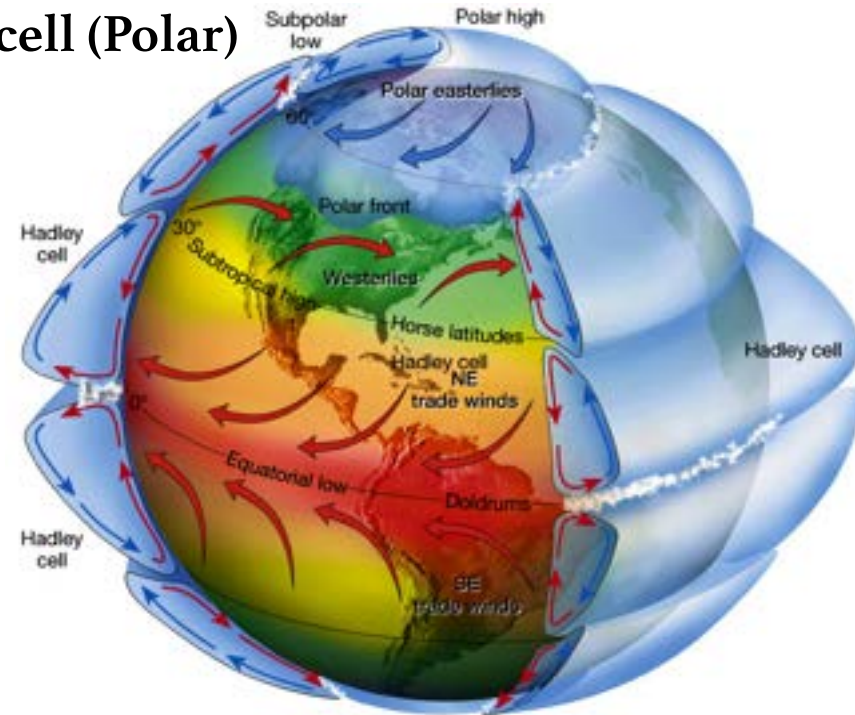
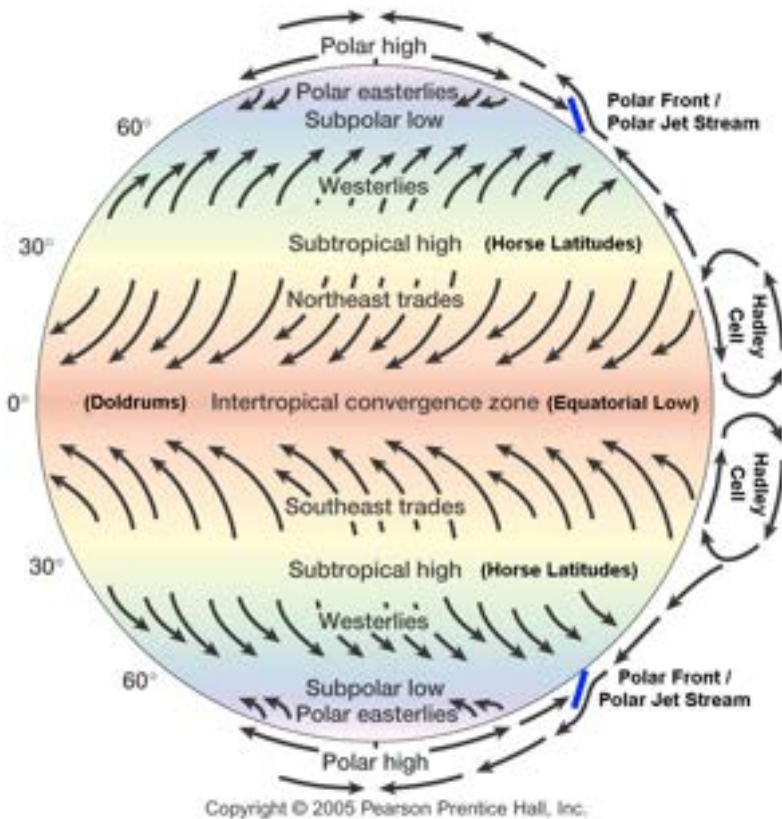
Characteristics of Hadley circulation:

- The Hadley circulation is a meridional circulation with an ascending branch in the tropics and a sinking branch in the subtropics.
- The ascending branch is associated with the zone of maximum solar heating.
- If the earth's surface was uniform, the mean position of the ascending branch would be at the Equator. Due to the asymmetric distribution of land between the Northern and Southern Hemispheres, and the very different thermal properties of land versus water, the mean position of the ascending branch of the Hadley cell is at about 5°N (5°N is often referred to as the Meteorological Equator).
- The ascending branch varies from about 5°S to 15°N over the course of the year.
- In the ascending branch, heat (primarily latent) is transported from the surface to the upper troposphere, where it is then transported poleward.
- If the earth were not rotating the Hadley circulation would be expected to reach all the way to the poles.



General Circulation-Three Cell Circulation

- Hadley cell (Tropical Cell)
- Ferrel cell (Midlatitude cell)
- Polar cell (Polar)



Hadley cell (Tropical Cell)

- The largest and deepest cell extend from the equator to between 30 and 40 degrees north and south, and are named Hadley cells, after English meteorologist George Hadley.
- Within the Hadley cells, the trade winds blow towards the equator, then ascend near the equator which forms the Inter-Tropical-Convergence Zone (ITCZ).
- From the top, the air flows towards higher latitudes, where it sinks to produce high-pressure regions over the subtropical regions.

Ferrel cell (Midlatitude cell)

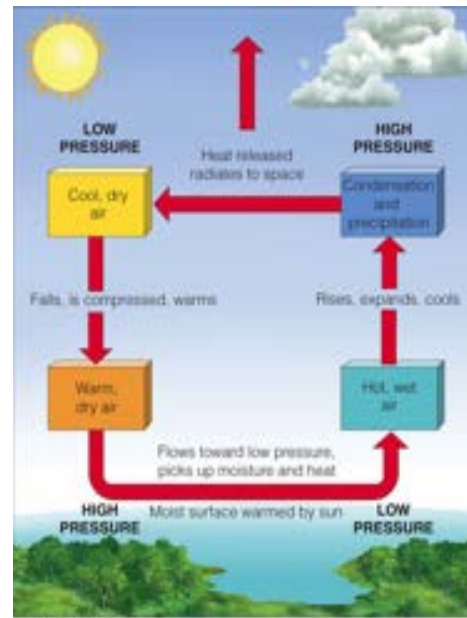
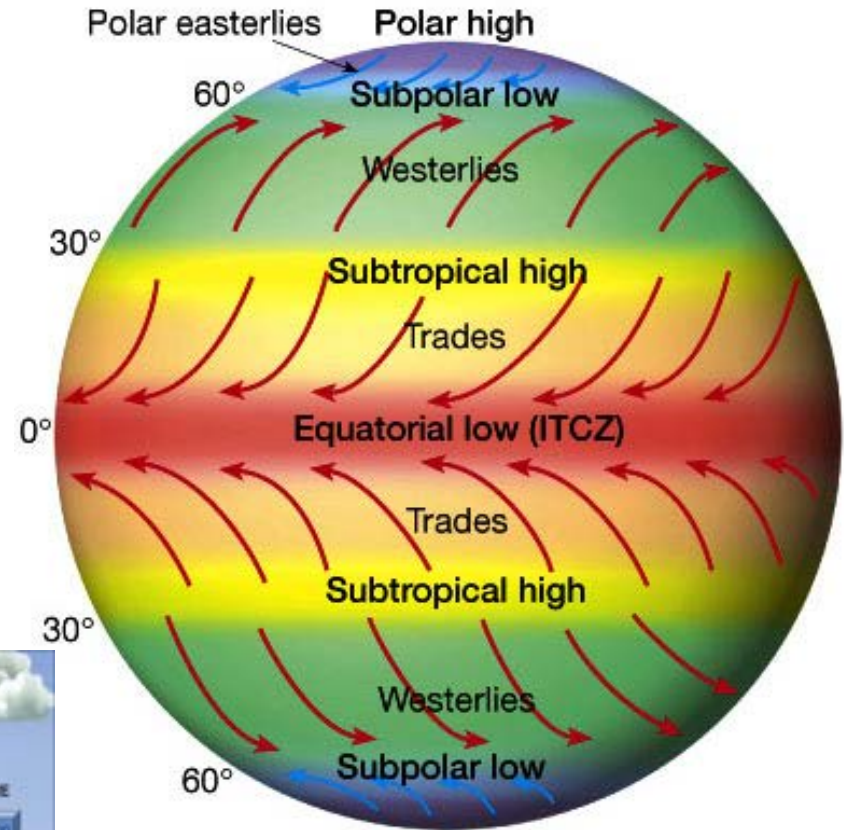
- In the Ferrel cell, air converges at low altitudes (30) to ascend along the boundaries between cool polar air and the warm subtropical air that generally occurs between 60 and 70 degrees north and south.
- The circulation within the Ferrel cell is complicated by a return flow of air at high altitudes towards the tropics, where it joins sinking air from the Hadley cell.
- The Ferrel cell moves in the opposite direction to the two other cells (Hadley cell and Polar cell) and acts rather like a **gear**.

Polar cell (Polar)

- The smallest and weakest cells are the Polar cells, which extend from between 60 and 70 degrees north and south, to the poles.
- Air in these cells sinks over the highest latitudes and flows out towards the lower latitudes at the surface.

Components of general circulation

- ❖ Intertropical Convergence Zone
- ❖ Trade Winds
- ❖ The Westerlies
- ❖ Subtropical **Highs**
- ❖ Subpolar **Lows**
- ❖ Polar Easterlies
- ❖ Polar Highs



Low vs. High Pressure

Components of the Three Cell circulations:

➤ **Equatorial Low or Inter-Tropical Convergence Zone (ITCZ)**

Warm air rises near the equator where trades meet; cloudy, rainy
ITCZ shifts with movement of sun, north in our summer, south in winter.

➤ **Trade Winds** Air flowing from subtropical highs towards ITCZ deflects by Coriolis forces into NE trades and SE trades. Trade winds are the most consistent winds in the atmosphere.

➤ **Westerlies** Air flowing poleward from subtropical highs are deflected by Coriolis force into westerlies at mid-latitudes.

➤ **Subtropical Highs** occur where rising tropical air at the equator moves poleward, cooling and sinking creating warm, dry conditions.

➤ **Polar Highs** occur at the two polar regions as cold air sinks and spreads equatorward. Coriolis deflection turns these high latitude winds into polar easterlies.

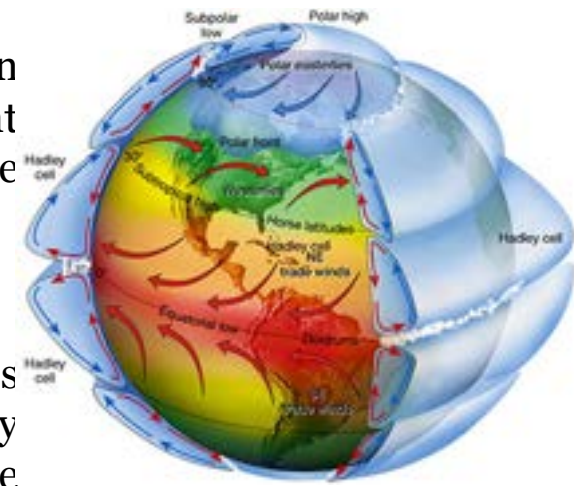
➤ **Polar front** Where cold polar easterlies meet warmer westerlies, low pressure and rising air forms the polar front, a zone of traveling cyclones.

Trade wind belts:

In the tropics, on both sides of the equator, lies a wide region where winds blow from east to west (easterlies) with a slight equatorward tilt. This region is named the trade wind belt, because of the steadiness of the air flow here.

Intertropical Convergence Zone (ITCZ):

The trade winds from the Northern and Southern Hemispheres converge into a narrow belt close to the equator, nowadays generally referred to as the Intertropical Convergence Zone (ITCZ). The convergence of the trade winds results in rising motion of the colliding air masses (to obey the law of mass continuity).



Midlatitude westerlies: North and south of the trade wind belt (in the Northern and Southern Hemispheres, respectively) lie regions where winds tend to blow from west to east (westerlies), and are therefore referred to as the westerly wind belts. Here the winds are highly variable and unsteady, especially so during winter.

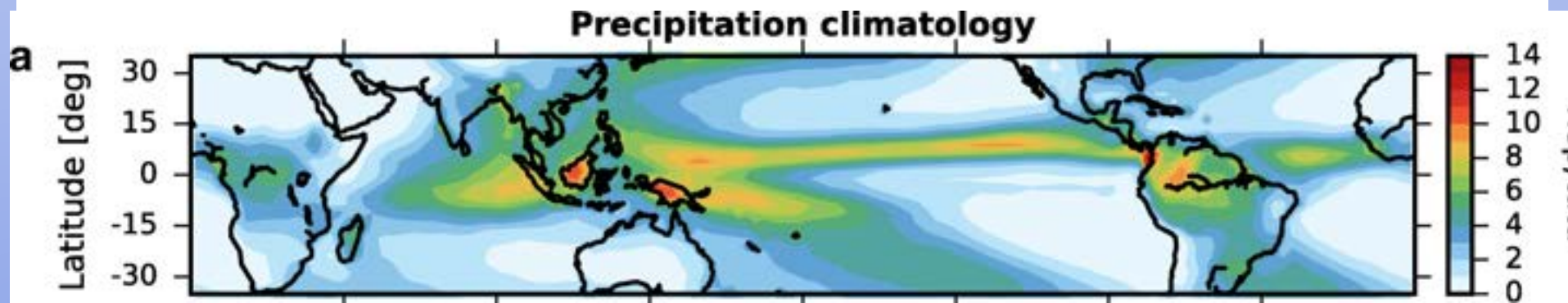
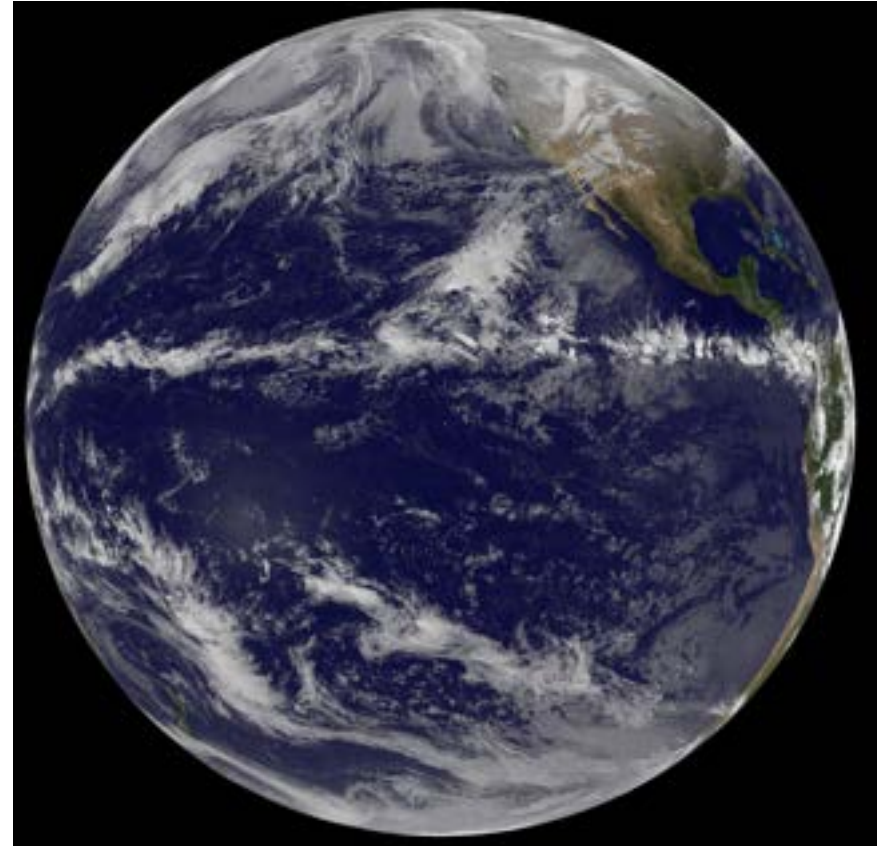
Subtropics: Between the trade wind regions lie the subtropics - regions of divergence and subsidence, where sunny weather with little clouds and no rain prevails. Most of the Earth's desert regions are found near 20-30° north and south of the equator.

Polar easterlies: Poleward from the westerly wind belt, winds with a generally easterly component prevail. The air here is cold, dry and stable, especially during winter, and is accompanied by subsidence from above.

Polar front: The convergence zone between polar easterlies and midlatitude westerlies is referred to as the polar front. It separates between the cold (and dry) polar air, and the relatively warm (and more humid) midlatitude air. The polar front can be thought of as the average expression of the transient frontal systems that move along with midlatitude cyclones.

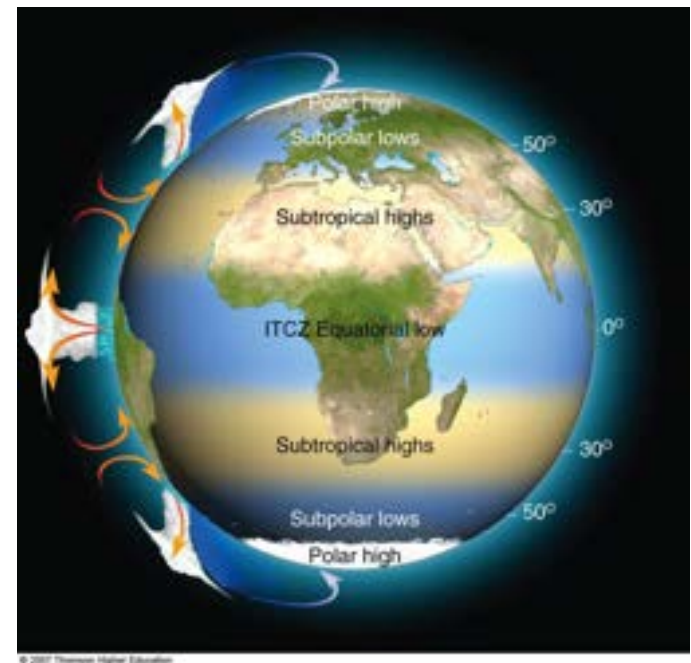
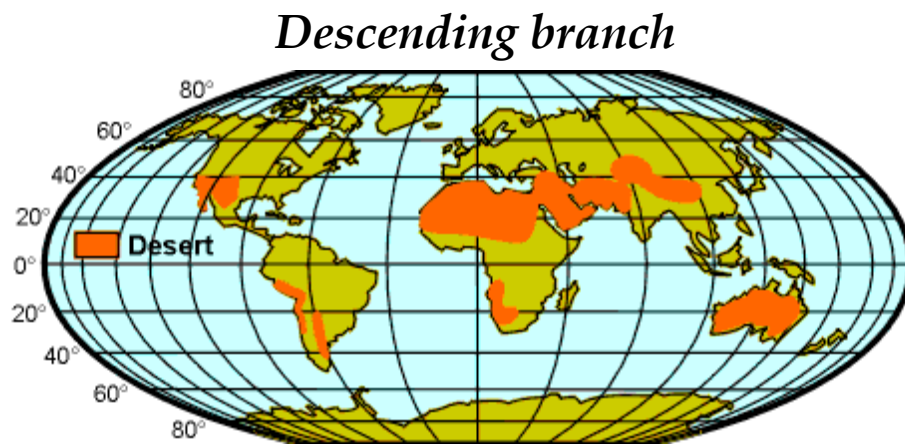
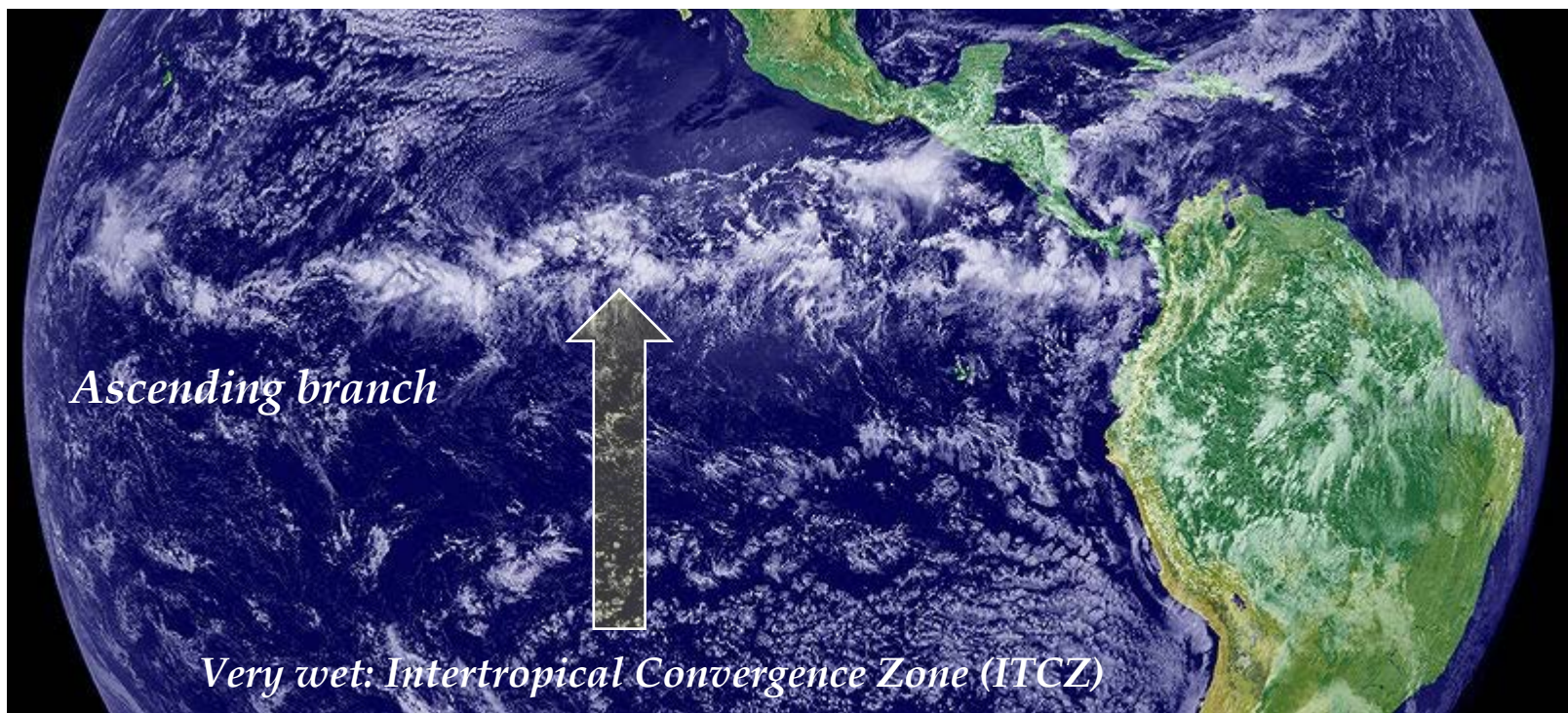
Intertropical convergence zone (ITCZ)

- Belt of low air pressure around equator
- Forms from surface heating
- Associated with clouds and rain
- Associated with convergence between the NE and SE trades
- Not necessarily on the equator (generally north of it)
- Also known as the doldrums

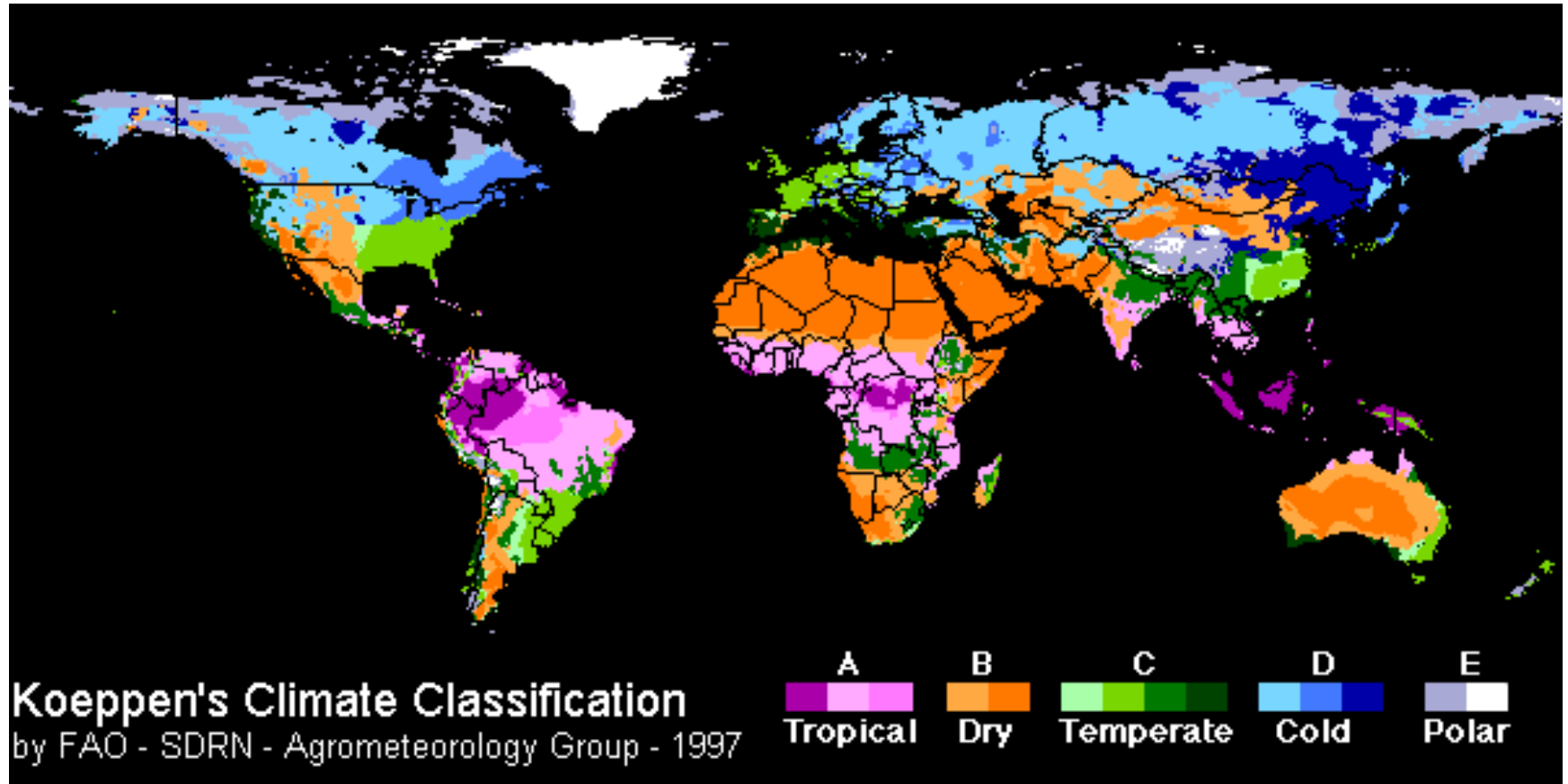


Typical Seasonal Positions of ITCZ



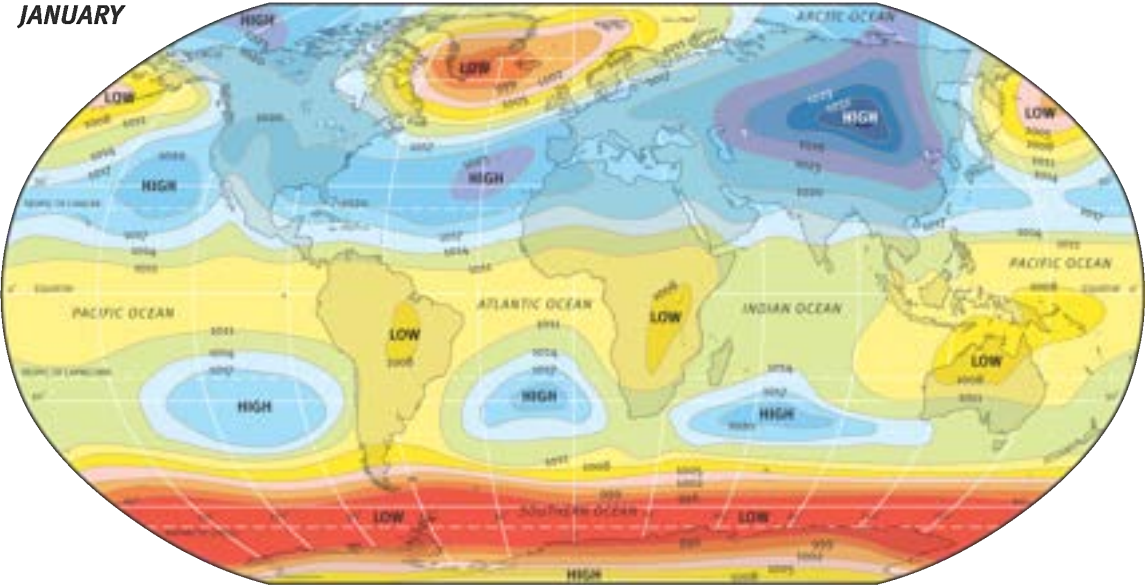


Climate Zones

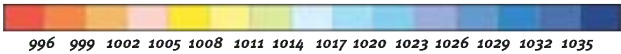


General Circulation Associated Pressure Belts

JANUARY

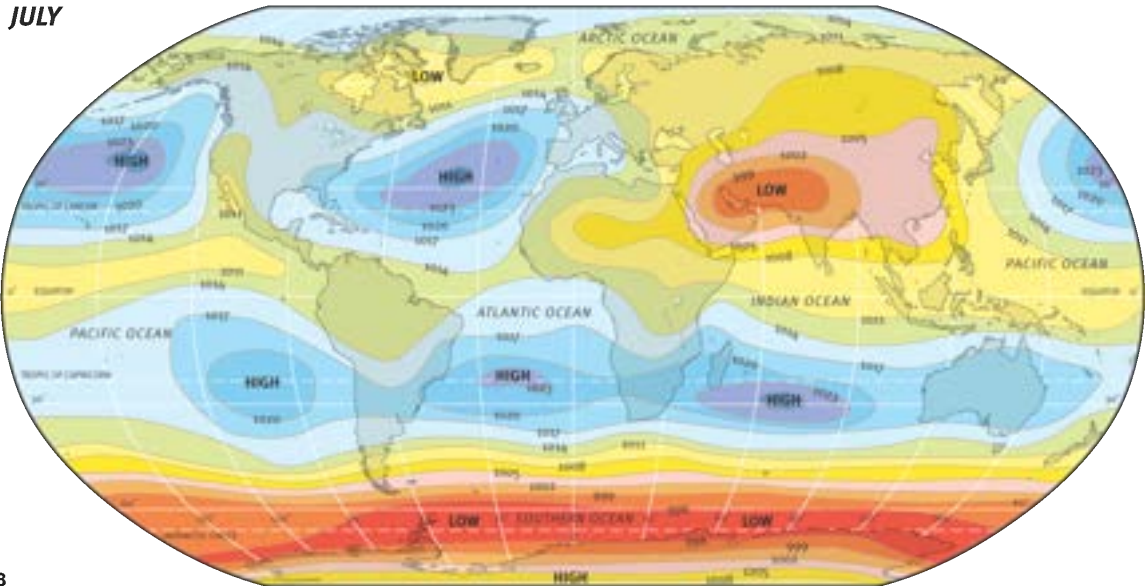


PRESSURE IN MILLIBARS



A

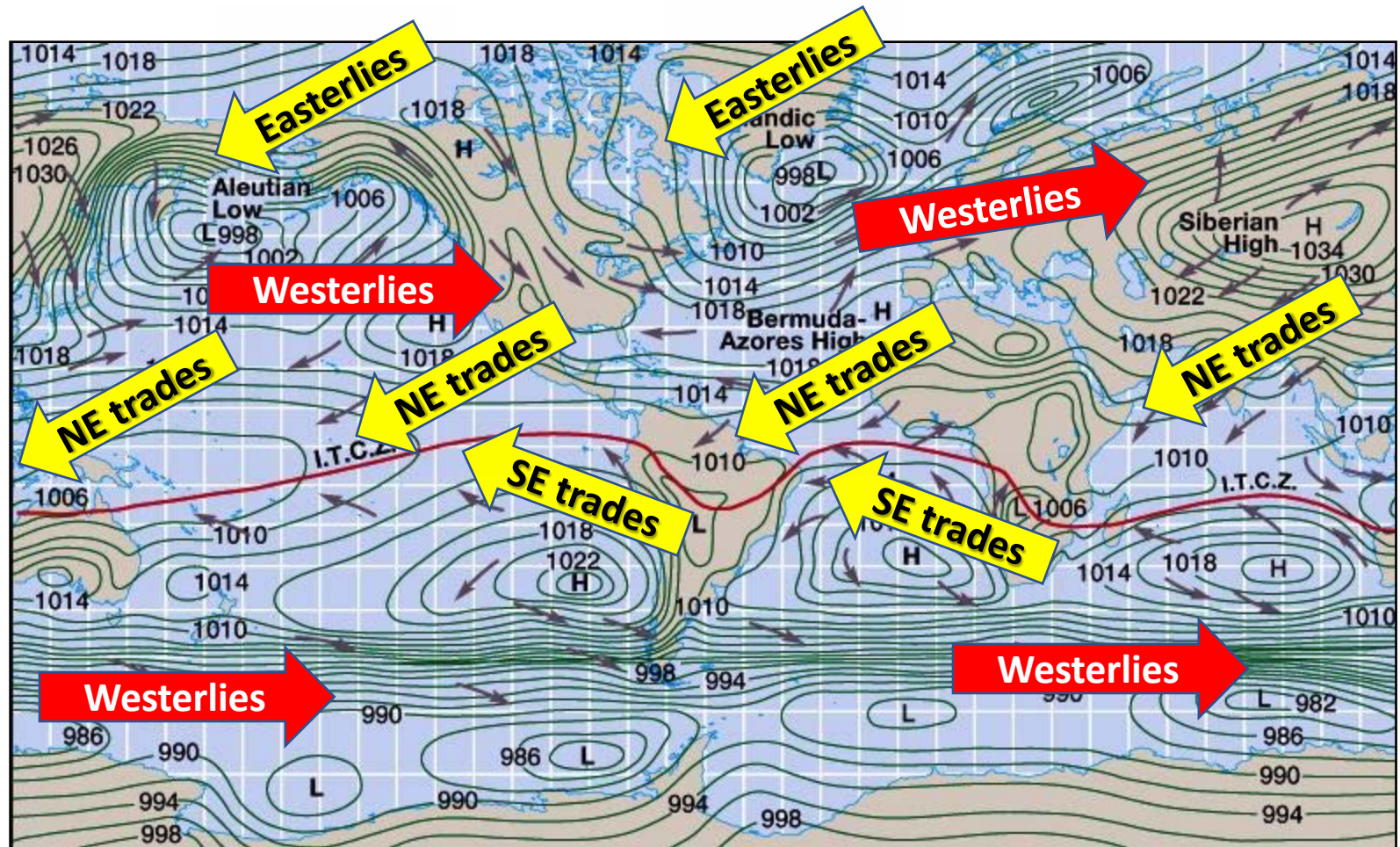
JULY



B

Global mean sea-level pressure (mb) patterns in January (A) and July (B).

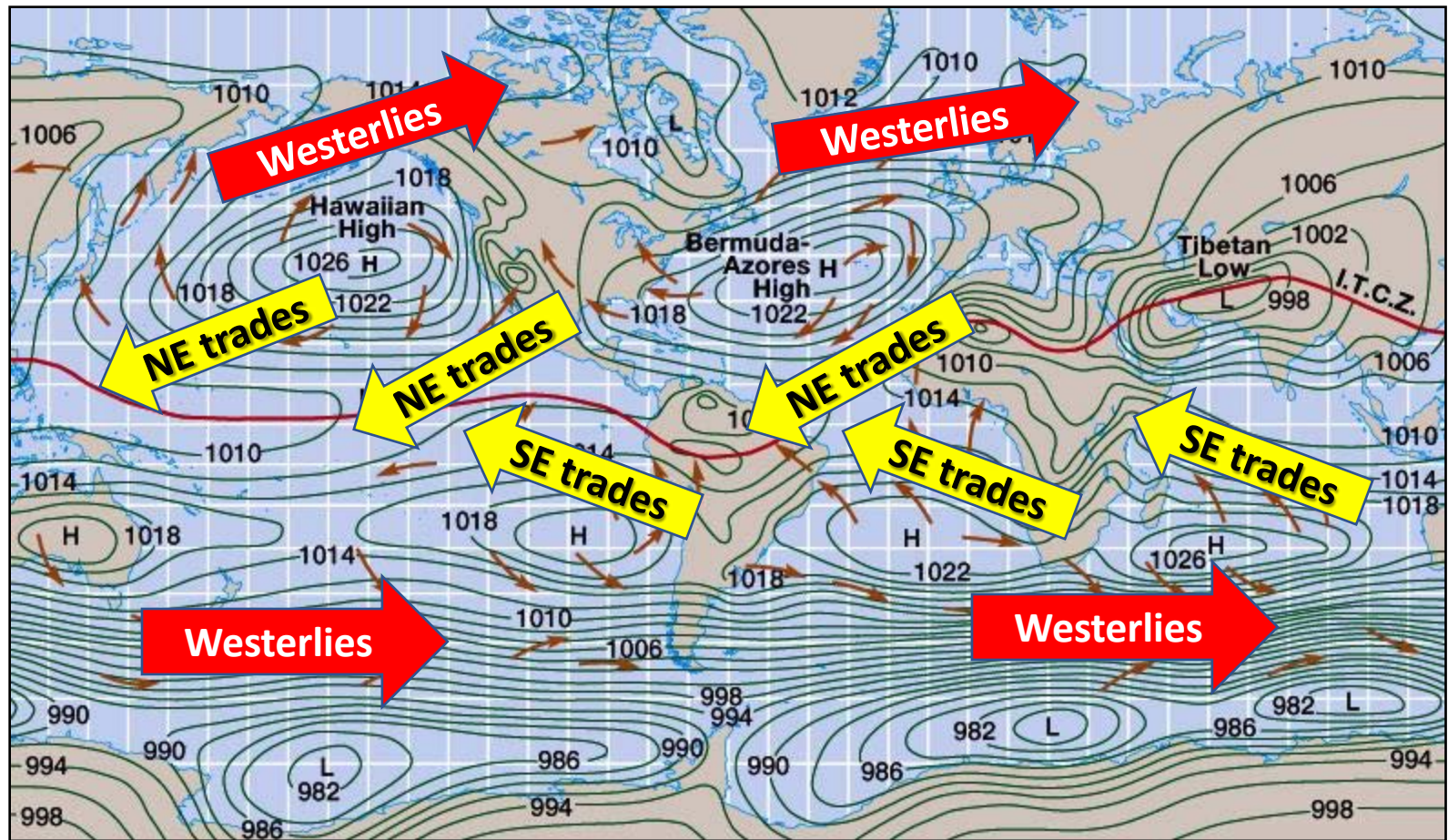
Circulation in a Global Context



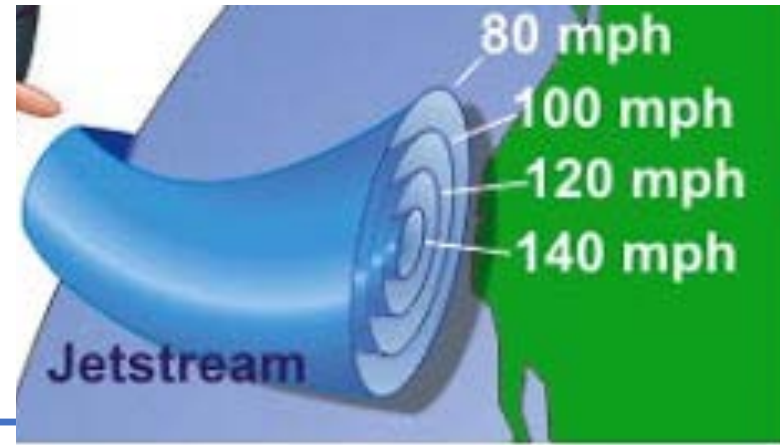
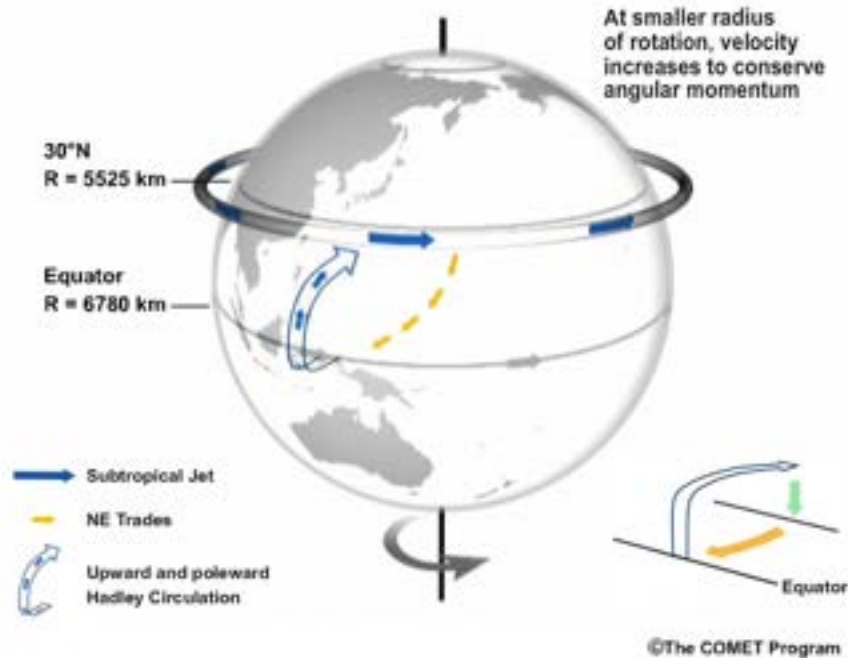
Mean surface pressure and winds during DJF (winter)

Mean surface pressure and winds during JJA (summer)

TRADE WINDS ARE EASTERLY WINDS



What will happen in the upper troposphere ?



Jet Stream:

Jet streams are narrow belts of high wind speeds that blow in the upper troposphere.

Jet stream is produced by thermal wind relation.

They form between areas of Low and High Pressure centres (interface of two different air).

It has the ability to separate the warm and cold air.

Important Jet Streams (strongest)

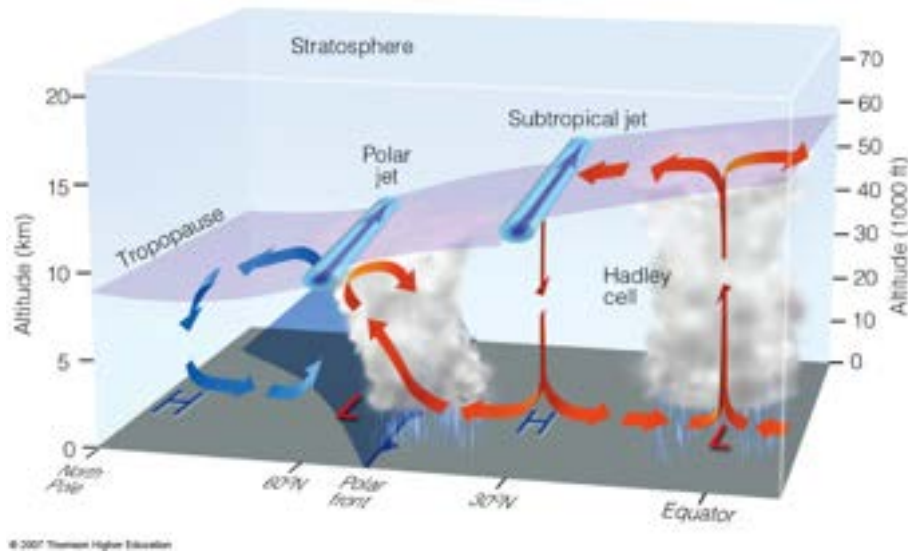
Subtropical jet stream

- The jet stream typically found between 20° and 30° latitude and at heights between 12 and 14 km.
- A jet stream is located at the poleward limit of the equatorial tropical air above the transition zone between tropical and mid-latitude air.

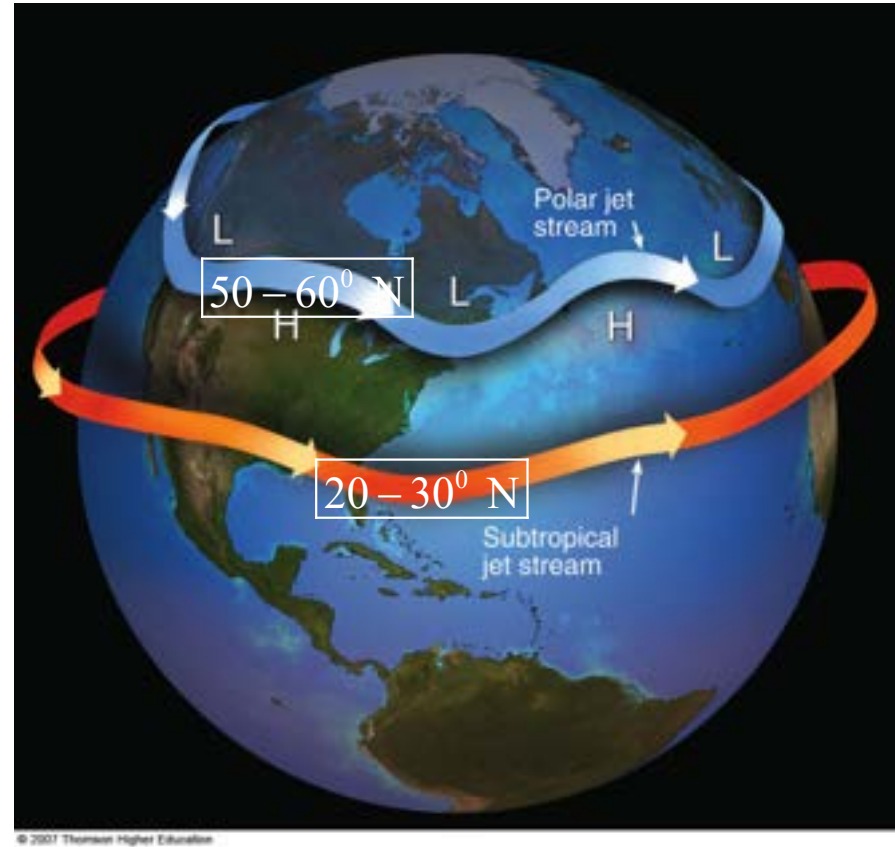
Polar jet stream (Polar jet)

- The jet stream that is associated with the polar front in middle and high latitudes. Usually located between 9 and 12 km.
 - Relatively strong winds concentrated within a narrow band in the atmosphere.
 - Jet stream core often exceeds 100 kts (more than 180 km/hr)
-
- Jet streams found at the upper levels at 200 hPa pressure level

Subtropical and polar jets:



Large temperature gradients at the surface correspond to strong winds aloft!



Subtropical Jet:

Located at the higher-latitude end of the Hadley Cell. The jet obtains its maximum wind speed (westerly) due the conservation of angular momentum (**of course as a consequence of thermal wind balance**)

Polar Jet:

Located at the thermal boundary between the tropical warm air and the polar cold air. The jet obtain its maximum wind speed (westerly) due the latitudinal thermal gradient (thermal wind relation).

Meridional heat transports:

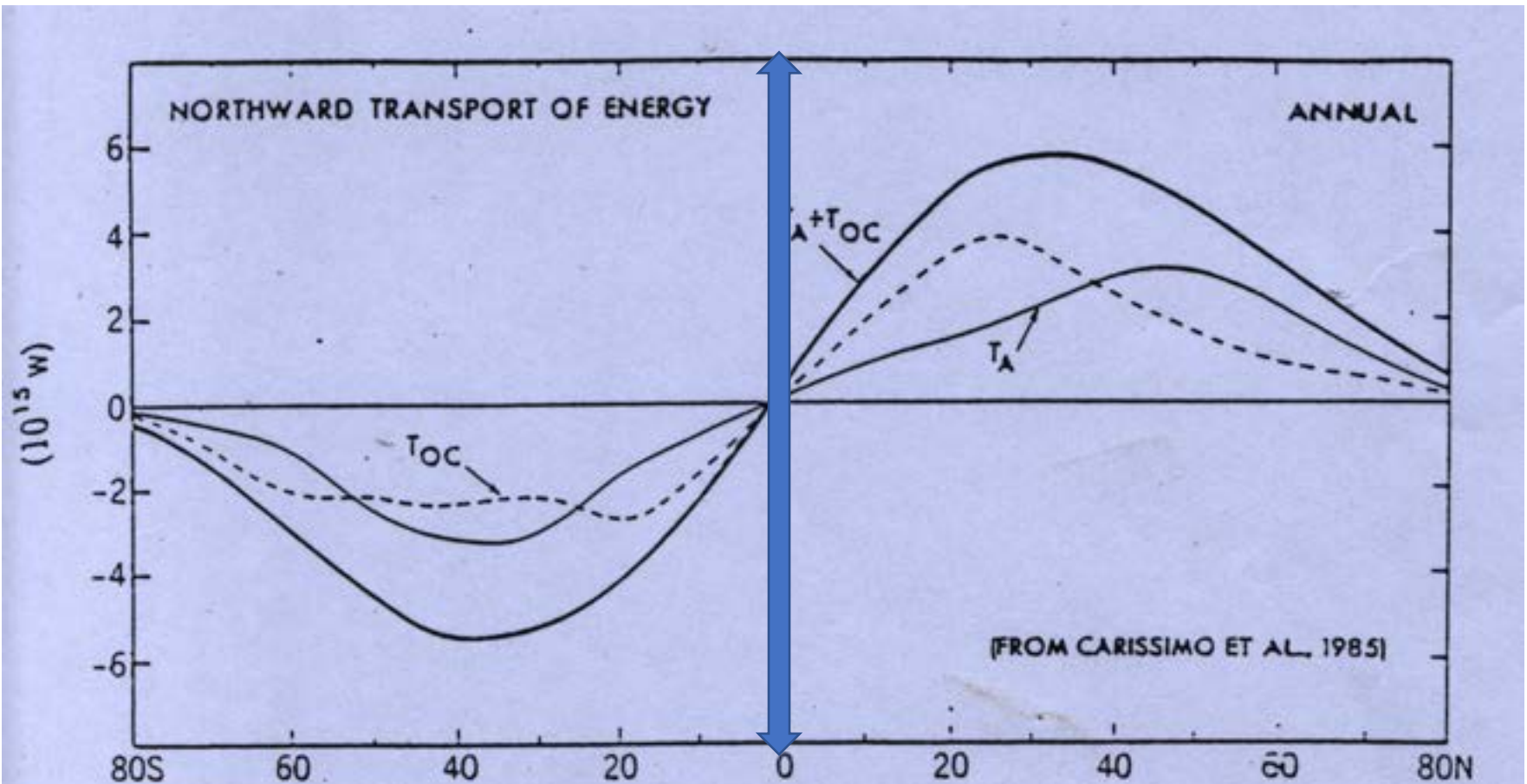
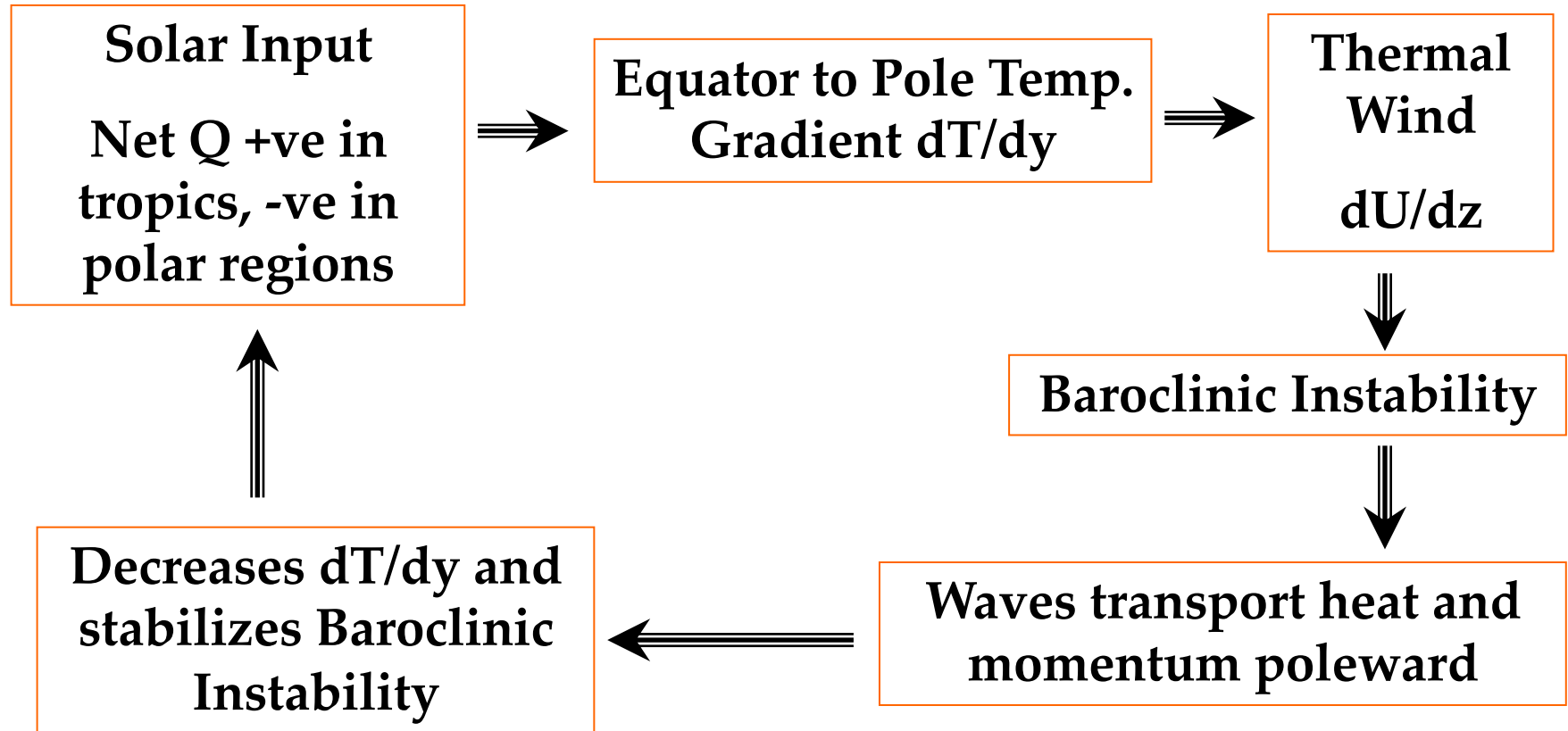


FIGURE 13.17. Meridional profiles of the northward transport of energy by the total system ($T_A + T_{OC}$), the atmosphere (T_A), and the oceans (T_{OC}) for annual-mean conditions in units of 10^{15} W (based on data from Carissimo *et al.*, 1985).

Maintenance of General Circulation of the Atmosphere

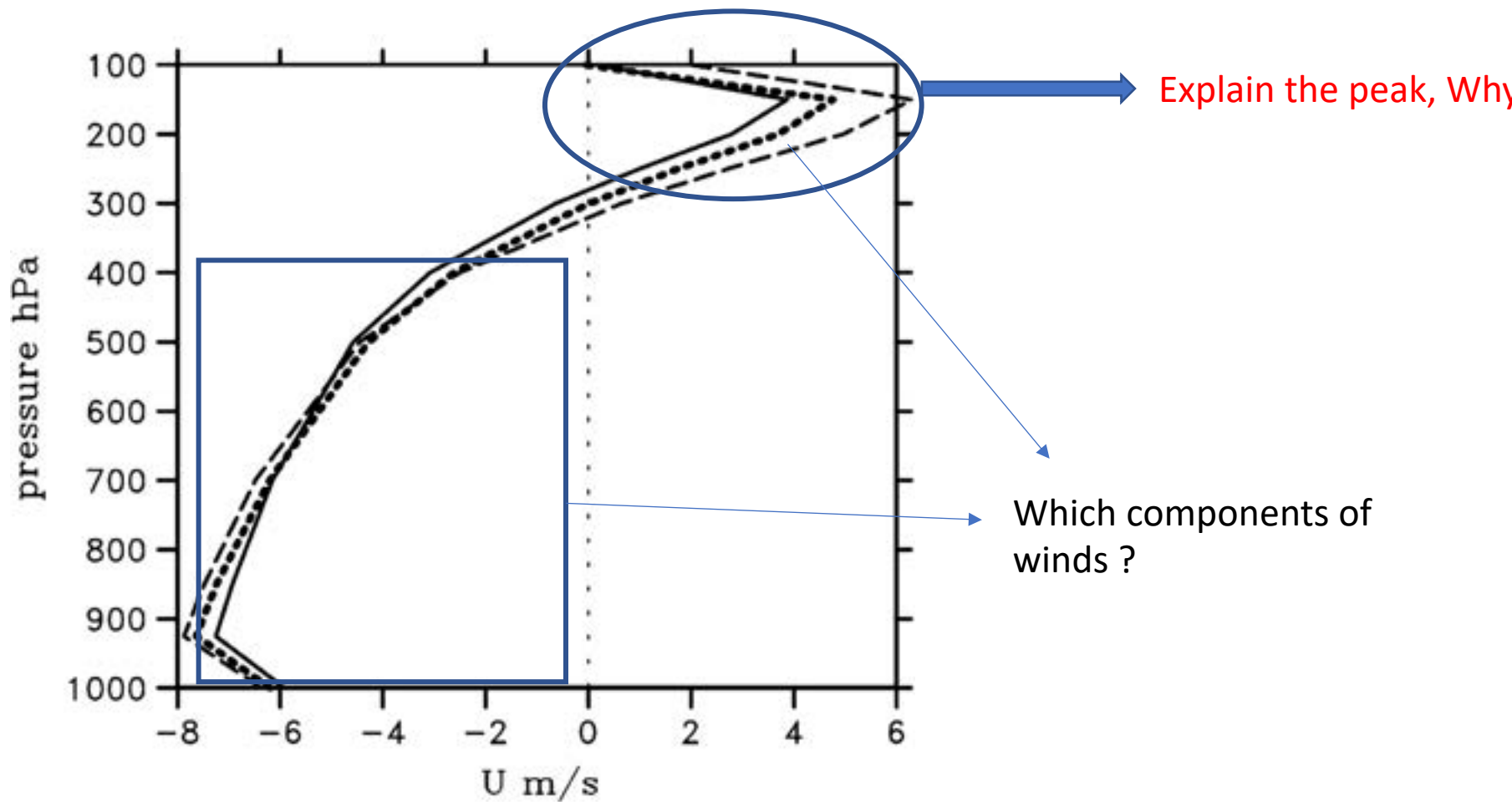


Baroclinic Instability: Instability that arises due to the existence of a meridional temperature gradient (vertical shear of the mean flow and a thermal wind).

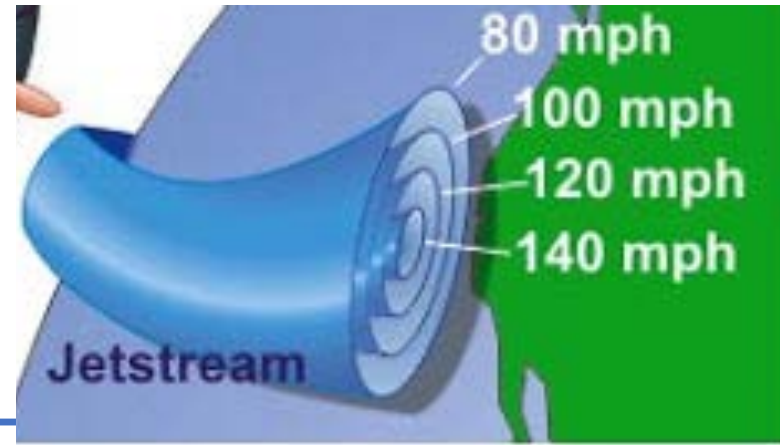
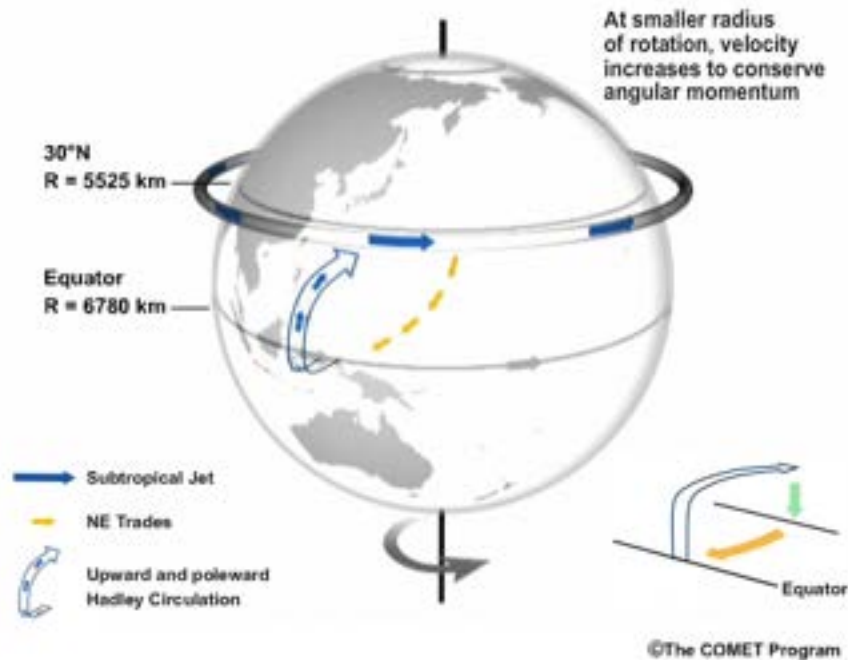
Some important features of the observed Mean condition of the atmosphere

- Surface **easterlies** in the tropics & surface **westerlies** in the middle latitudes
- **Westerly jet stream** in the upper atmosphere subtropics. Winter hemisphere jet tends to be stronger than the summer hemisphere one.
- **Easterly jet** in the upper atmosphere over the equatorial region during summer monsoon region
- Three cell meridional structure
- Equator to pole temperature difference is about 60°K in the winter hemisphere and about 35°K in the summer hemisphere
- The temperature gradient in the meridional direction is weak in the tropics and strong in the middle latitude.
- Height of the tropopause is much lower in the polar region as compared to the equatorial region

Vertical Structure of zonal wind circulation at 45° N



What will happen in the upper troposphere ?



Jet Stream:

Jet streams are narrow belts of high wind speeds that blow in the upper troposphere.

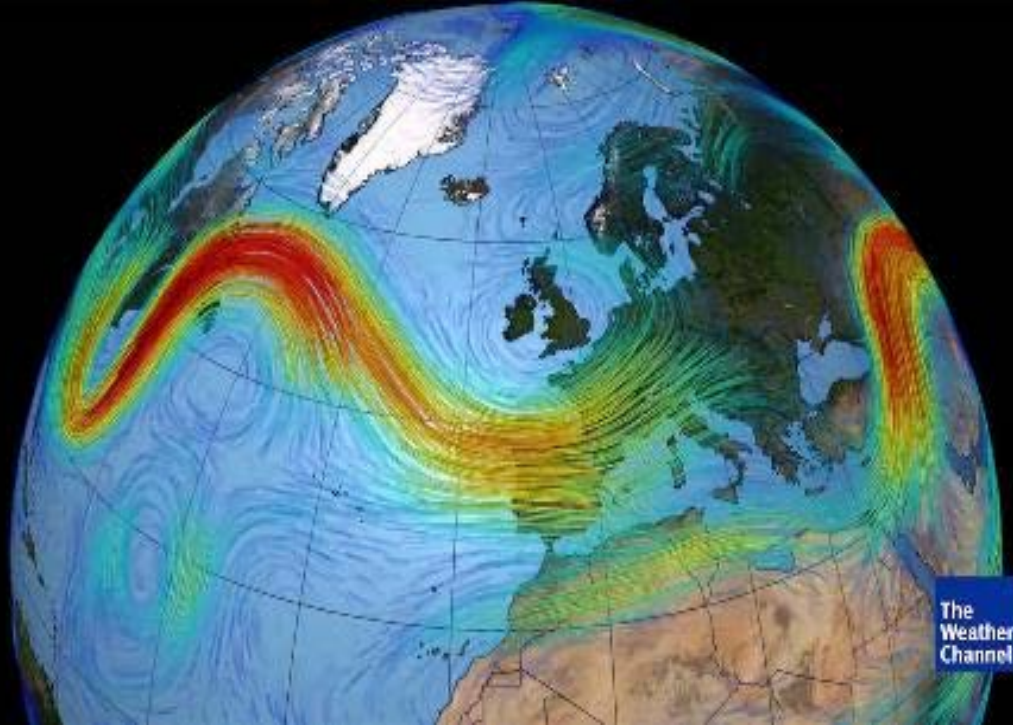
Jet stream is produced by thermal wind relation.

They form between areas of Low and High Pressure centres (interface of two different air).

It has the ability to separate the warm and cold air.

Why Do We Care About the Jet streams?

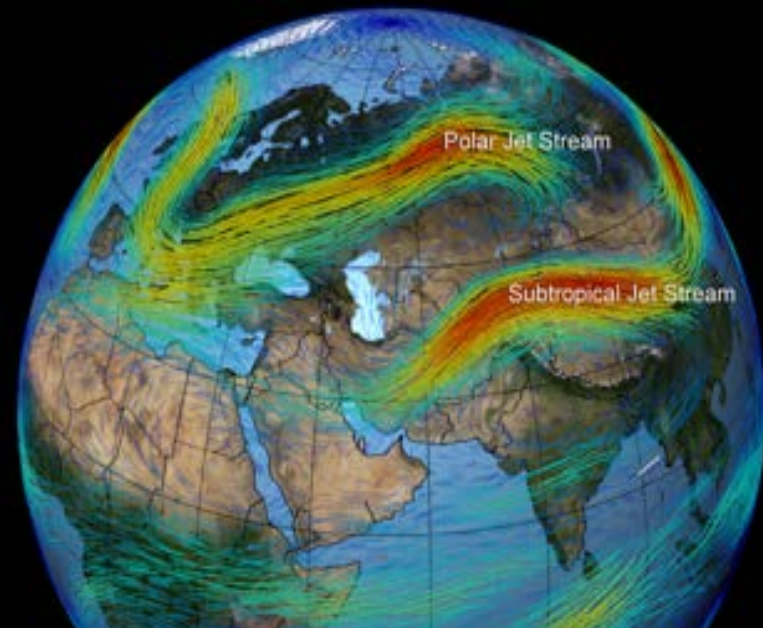
- Knowing the location of jet streams can aid in **weather forecasting**.
- The path of jet streams steers cyclonic systems at lower levels in the atmosphere.
- The main commercial relevance of the jet streams is in **air travel**, as flight time can be dramatically affected by either flying with the flow or against the flow of a jet stream.
- **Clear-air turbulence** is could be found in a jet stream's vicinity. It can cause aircraft to plunge and is a potential hazard to aircraft passenger safety.



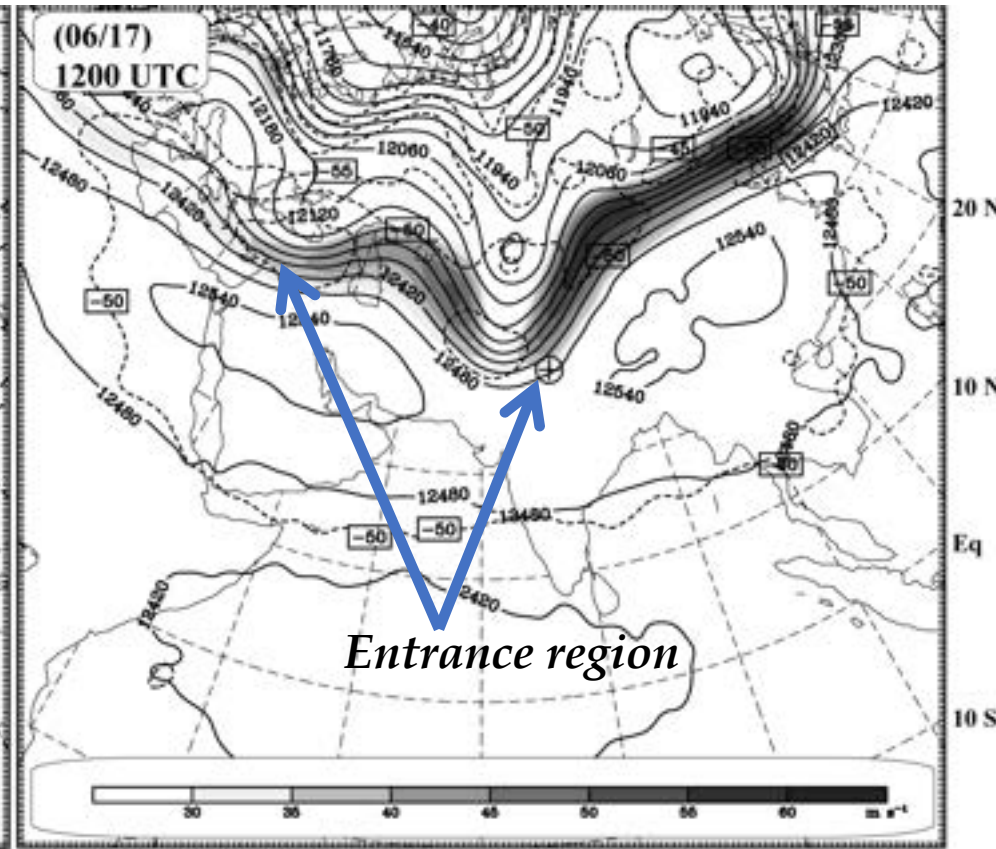
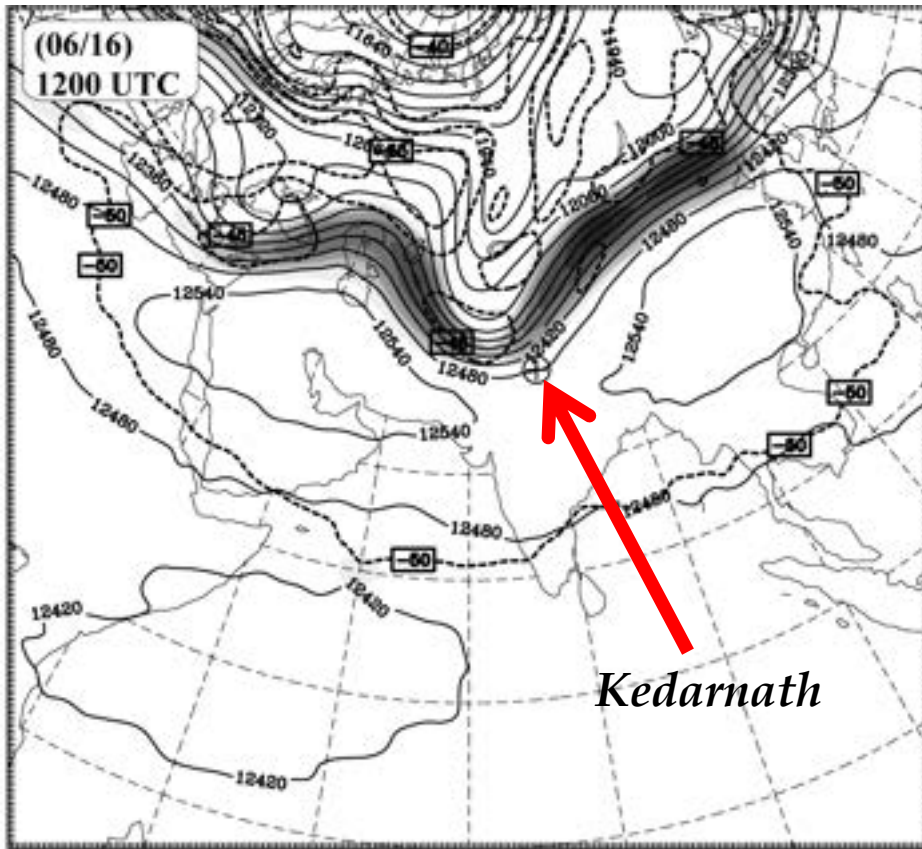
Normally the jet stream runs fairly directly from west to east and pushes weather systems through quite quickly.

However, sometimes the steering flow of the jet stream can meander (a wavy like), curving north and south as it heads east.

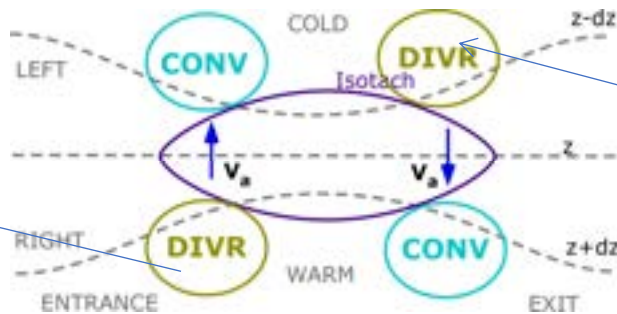
❖ The jet stream guides weather (or low pressure) systems which bears rain and unsettled conditions



200 hPa Jet stream – Uttarkhand flood



16-17 June 2013

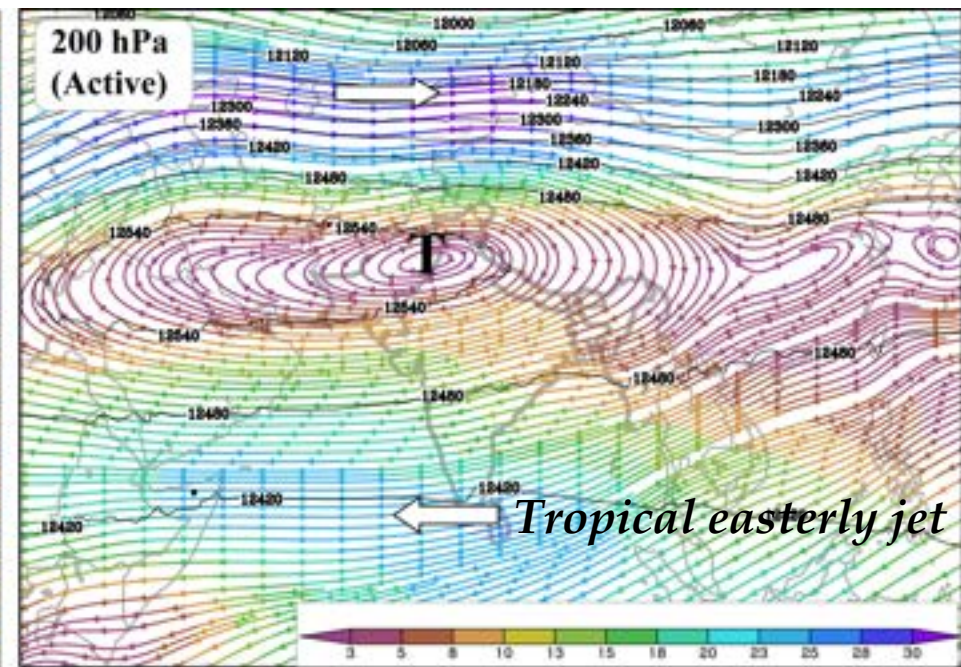
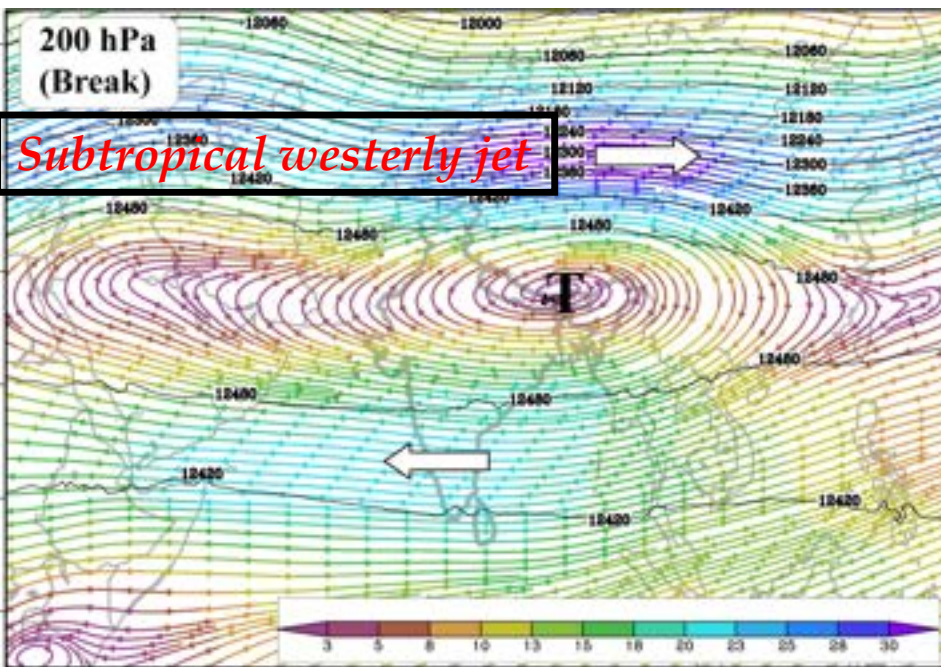


*Thermally direct
Circulation
(entrance region)*

*Thermally indirect
Circulation
(exit region)*



Subtropical jet during active and break spells of monsoon



Winds and geopotential height at 200 hPa

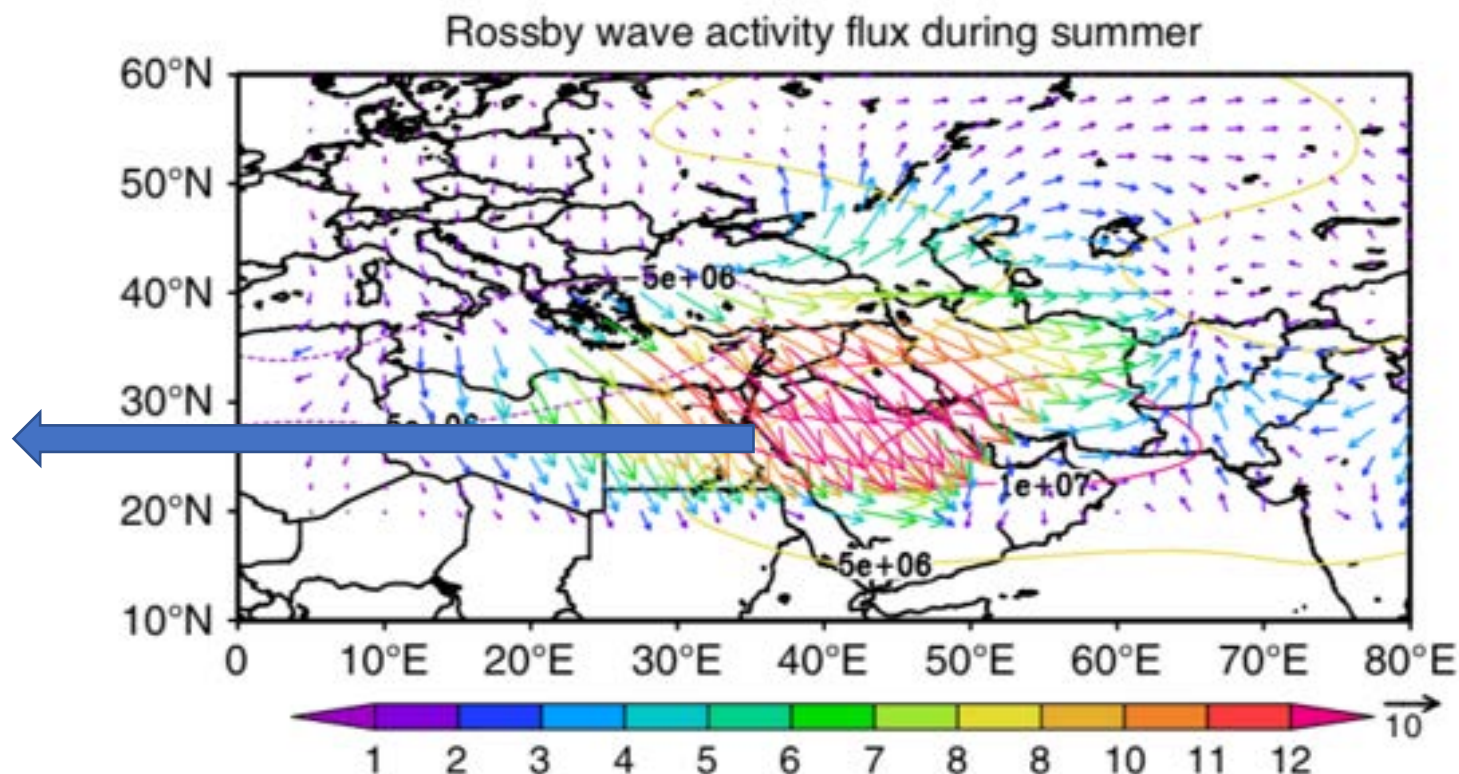
T = Tibetan anticyclone

RESEARCH ARTICLE

Prominent mode of summer surface air temperature variability and associated circulation anomalies over the Arabian Peninsula

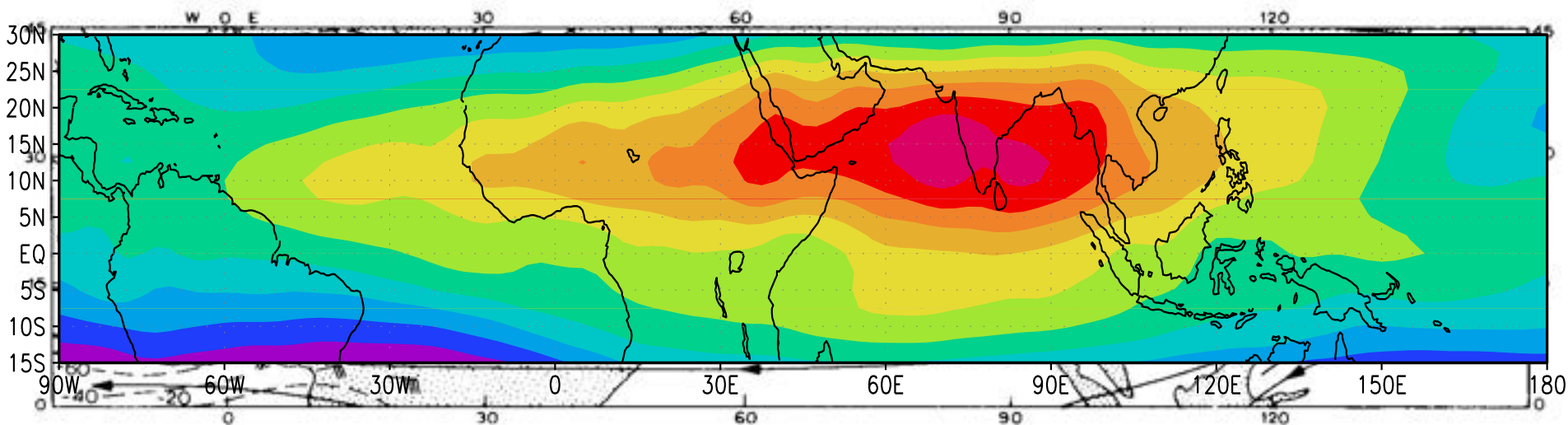
Raju Attada¹  | Ramesh Kumar Yadav² | Ravi Kumar Kunchala^{1,4}  | Hari Prasad Dasari¹ | Omar Knio³ | Ibrahim Hoteit¹

Leads to Strong
adiabatic
warming over
the Saudi
Arabia



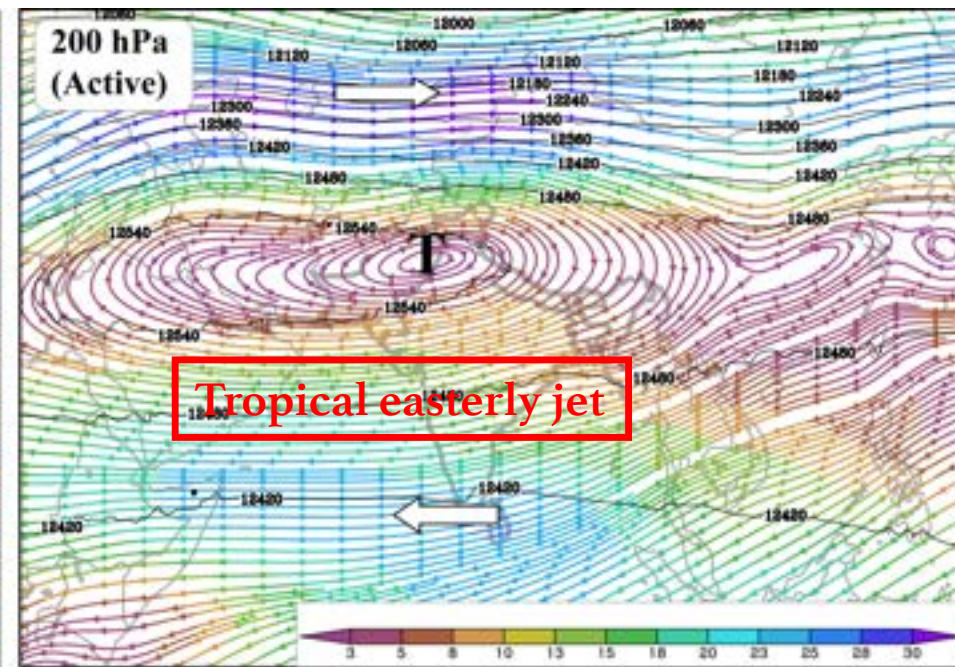
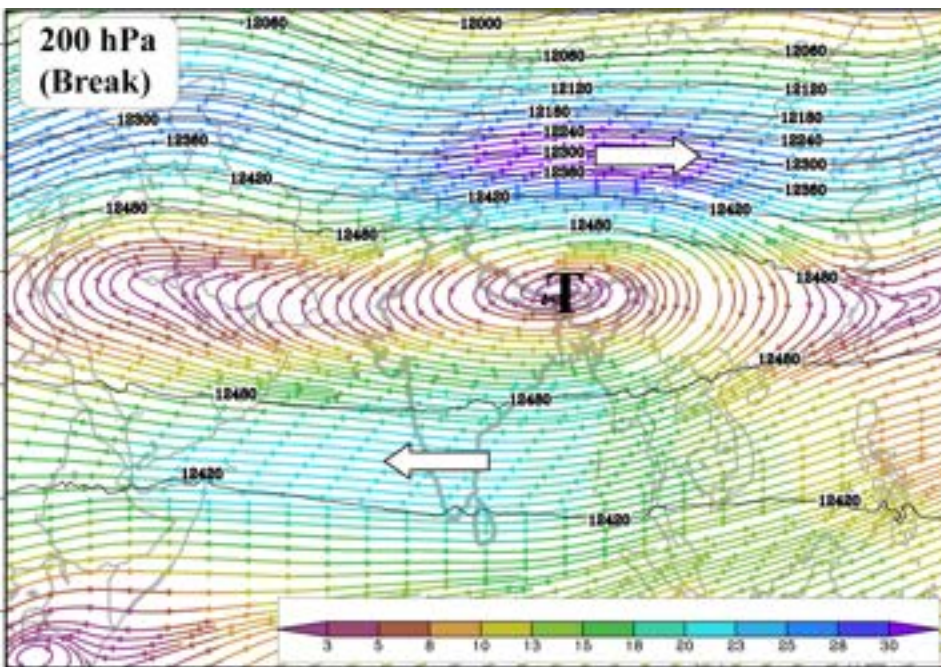
Regional Jet streams: Tropical Easterly Jet stream

- ✓ **The Tropical Easterly Jet** is an upper level easterly wind that starts in June and continues until September. This strong flow of air that develops in the upper atmosphere during the Asian monsoon is centred on 15°N , $50\text{--}80^{\circ}\text{E}$ and extends from South-East Asia to Africa.
- ✓ The strongest development of the jet is at about 15 km above the Earth's surface with wind speeds of up to 40 m/s over the Indian Ocean.
- ✓ Jet formed due to intense **heating over Asian (Tibet) plateau**.
- ✓ Formation: Thermal gradient is reversed, resultant wind is the tropical easterly jet.



- Found between 200 and 100 hPa pressure levels.
- Core of Tropical Easterly jet is
 - 15°N over India and
 - 5 °N over West Africa.
- Usually found in Summer.
- Feature: Band of easterly winds from equator to about 25° N.
- Jet speeds reached in only 2 zones.
 - Philippines to Sudan (max mean speed 100 Kt) and
 - Over W. Africa (max mean speed 60-70 Kt).

Tropical Easterly Jet during monsoon



Winds and geopotential height at 200 hPa

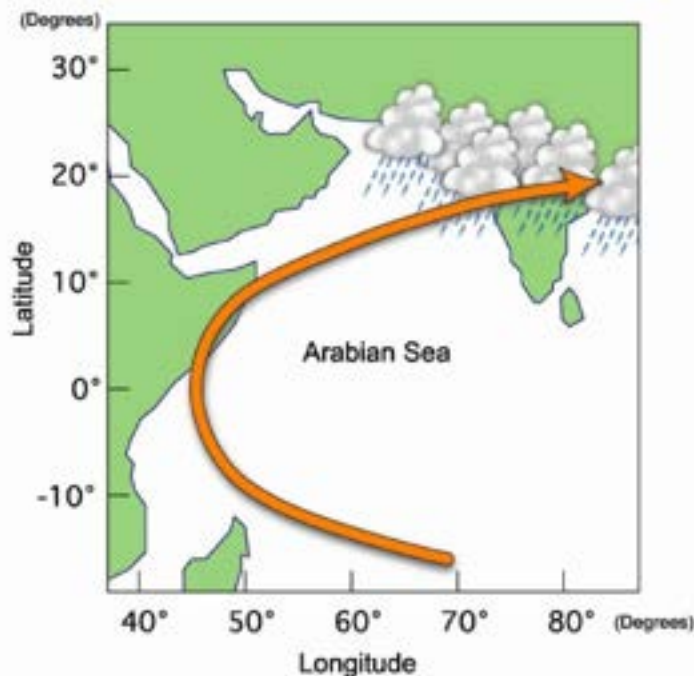
Weaker TEJ

T = Tibetan anticyclone

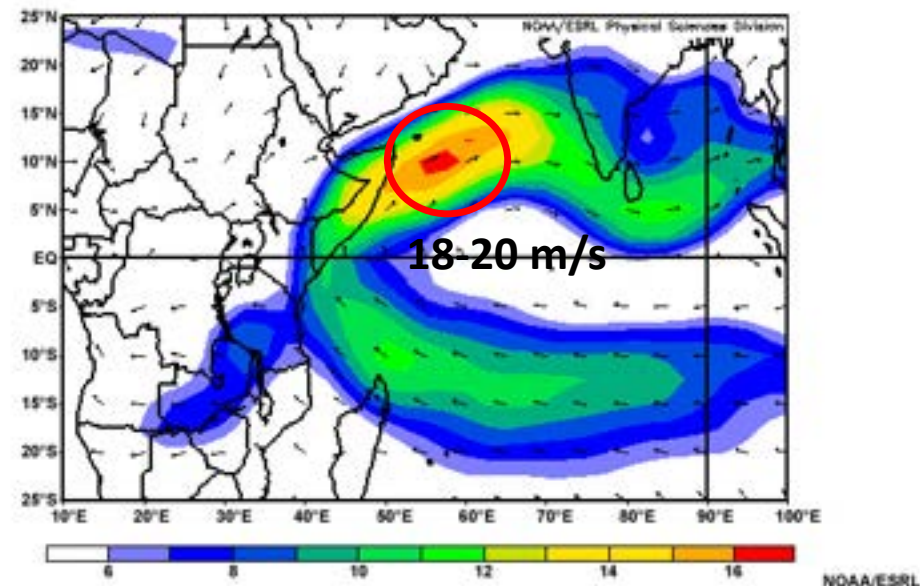
Stronger TEJ

Somali Jet or Low level Jet (Findlater)

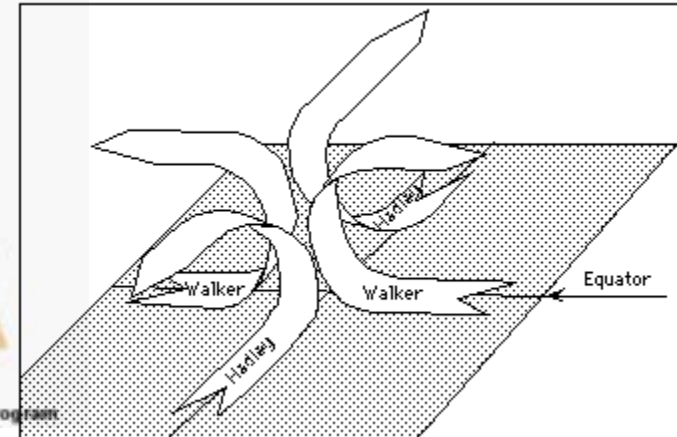
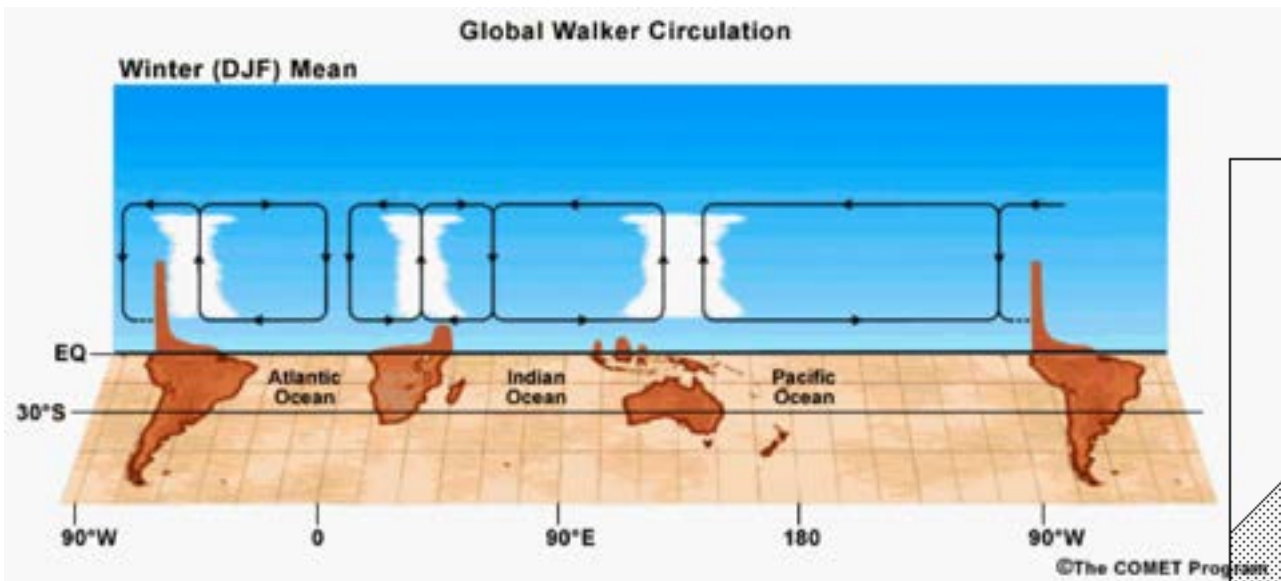
- The Somali jet is one of the most prominent wind systems of the lower troposphere during the Asian/Indian summer monsoon.
- Its strongest winds are found during the months of June-July-August at a height of roughly 1 to 1.5 km above the Earth's surface off the coast of Somalia and extending over the northern Arabian Sea.
- Max speeds at the core of LLJ



Jul-Sep Mean 850-hPa Wind Velocity (m s^{-1}), NCEP/NCAR Reanalysis, 1979-2009



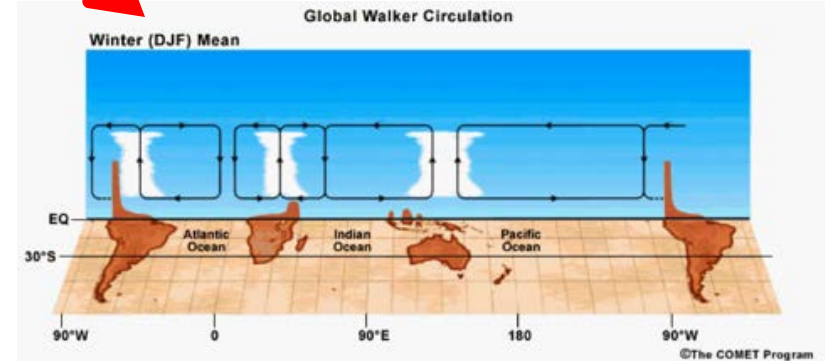
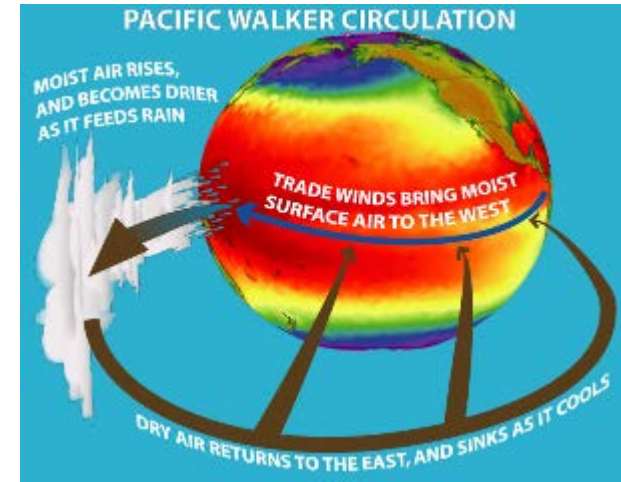
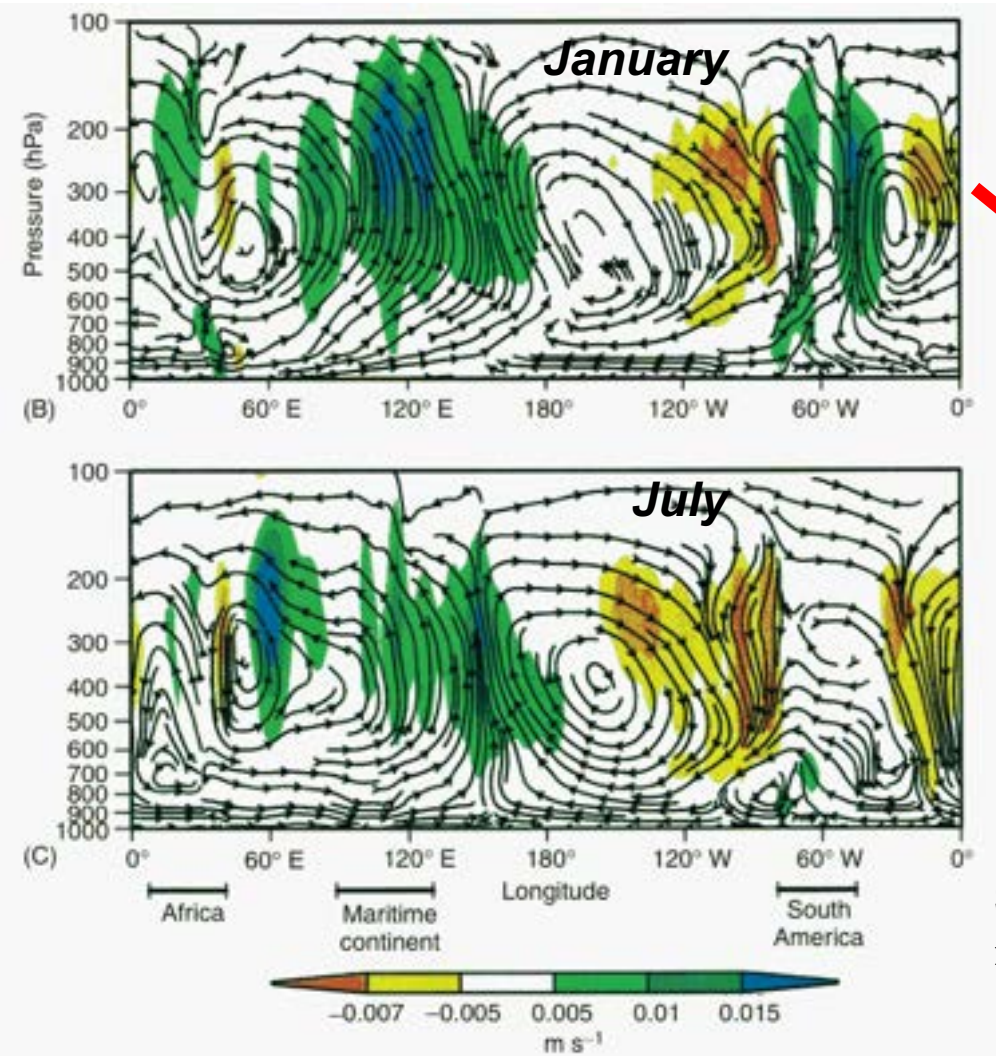
Walker circulation (Zonal Cell)



- The Walker circulation is a 2D circulation (large-scale zonal circulation) along the equator in the longitude-height plane.
- This circulation consists of cells comprising rising motions in major centres of equatorial convection (the Amazon, Africa, western Pacific) and sinking motions over adjacent ocean basins.

- The Walker circulation **is the result of a difference in surface pressure and temperature** (is caused by differences in heat distribution between ocean and land) over the western and eastern tropical Pacific Ocean.
- The Walker circulation is caused by the pressure gradient force that results from a high pressure system over the eastern Pacific ocean, and a low pressure system over Indonesia.
- Walker circulation has been slowing since the mid-19th Century due to global warming

Walker circulation in the Data



- Areas of strong ascent/ descent are shaded
- Some seasonality is evident, e.g., the strong South America/Atlantic cell in January is absent in July.

Air Masses

- An air mass is an extremely large body of air whose physical properties are fairly similar over horizontal distances of thousands of kilometres in any horizontal direction at any given altitude.

Physical Properties: Temperature

Moisture content

Density

Lapse rate

- Properties and degree of uniformity depend on:

Source of the air (i.e ocean or land)

Age of air mass

- Originate in high-pressure regions, and move towards low pressure
- Temperature and moisture conditions determined by SOURCE REGION of air mass

- How they trigger:-

If a large body of air sits over an area of land or water for a long period of time, it will take on the characteristics of the land or water beneath it.

Basic nature of the Air masses

- Air masses over the equator will have high temperatures.
 - Air masses over polar regions will have low temperatures.
 - Air masses over water (maritime) will have high humidity (moisture content).
 - Air masses over land (continental) will have low humidity (moisture content).
-
- Air mass forms over the ocean and has humid air.
 - Air mass forms over land and has dry air.
 - Air mass forms in the tropics. A tropical air mass has warm air.
 - Air mass forms near the poles. A polar air mass has cold air.
-
- Maritime - forming over ocean surfaces - (m) which create relatively humid air masses
 - Continental – forming over land surfaces - (c), which create relatively dry air masses

Sources & Classification

Air Mass Temperature Classifications:

Temperature of an air mass is classified by the general characteristics of its source region

A = Arctic

P = Polar

T = Tropical



Airmass moisture characteristics:

m = maritime (ocean) surfaces

c = continental (land) surfaces

The combination of temperature and moisture gives us basic air mass types



Nomenclature of Air Masses

<u>Name</u>	<u>Symbol</u>	<u>Source Region</u>
Arctic/Antarctic	A/AA	Arctic/Antractic Ocean
Polar	P	50-60N
Tropical	T	20-35N
Equatorial	E	<10N
Maritime	m	Oceans
Continental	c	Continents

- Air masses that form over water are called **maritime**.
- Air masses that form over land are called **continental**.

mT – maritime tropical
cT – continental tropical
mP – maritime polar
cP – continental polar
cA – continental arctic

Classification

- **Maritime tropical:** warm, humid weather.
- **Continental tropical:** hot, dry weather.
- **Maritime polar and maritime arctic:** pick up moisture as they travel across oceans and tend to produce cool, moist weather.
- **Continental polar (cP) and continental arctic (cA):** usually produce:
 - a. *winter: cold, dry weather*
 - b. *summer: cool, pleasant weather*

• **TABLE 11.1**

Air Mass Classification and Characteristics		
SOURCE REGION	POLAR (P)	TROPICAL (T)
Land continental (c)	cP Cold, dry, stable	cT Hot, dry, stable air aloft; unstable surface air
Water maritime (m)	mP Cool, moist, unstable	mT Warm, moist; usually un- stable

Maritime Tropical (mT)

- Very warm and very humid – develop in tropical and sub-tropical latitudes
 - Originate over warm waters of Ocean
 - Can form year around, but are most common in summer
 - Important source of moisture feeding storms all year round
-
- **Responsible for hot, humid days of summer** across southern Europe and much of the south and eastern half of North America



Continental Tropical (cT)

- Very warm; develop in lower subtropical latitudes
- Very dry because of formation over land
- form over the deserts of North Africa and southwest Asia; bring heat to southern Europe during summer
- as air mass moves eastward, moisture is evaporated into it, making it more mT
- these air masses rarely form in winter



Maritime Polar (mP)

- Very cool and moist; typically bring cloudy, damp weather
- Form over northern Pacific and northern Atlantic Oceans
- Most often influence the British Isles and Scandinavia.
- Maritime polar air masses can form any time of the year and are usually not **as cold as continental polar air masses**.



Continental Polar (cP)

- Very cold, having developed over sub-polar regions (**not as cold as Arctic air masses**)
- very dry, due to cold and developing over **land**
- Form farther south in the subpolar Siberia and eastern continental Europe
- common across continent during winter
- Do form in summer, but mostly only in interior Siberia and Russia
- **Typically bring clear and pleasant weather during the summer**

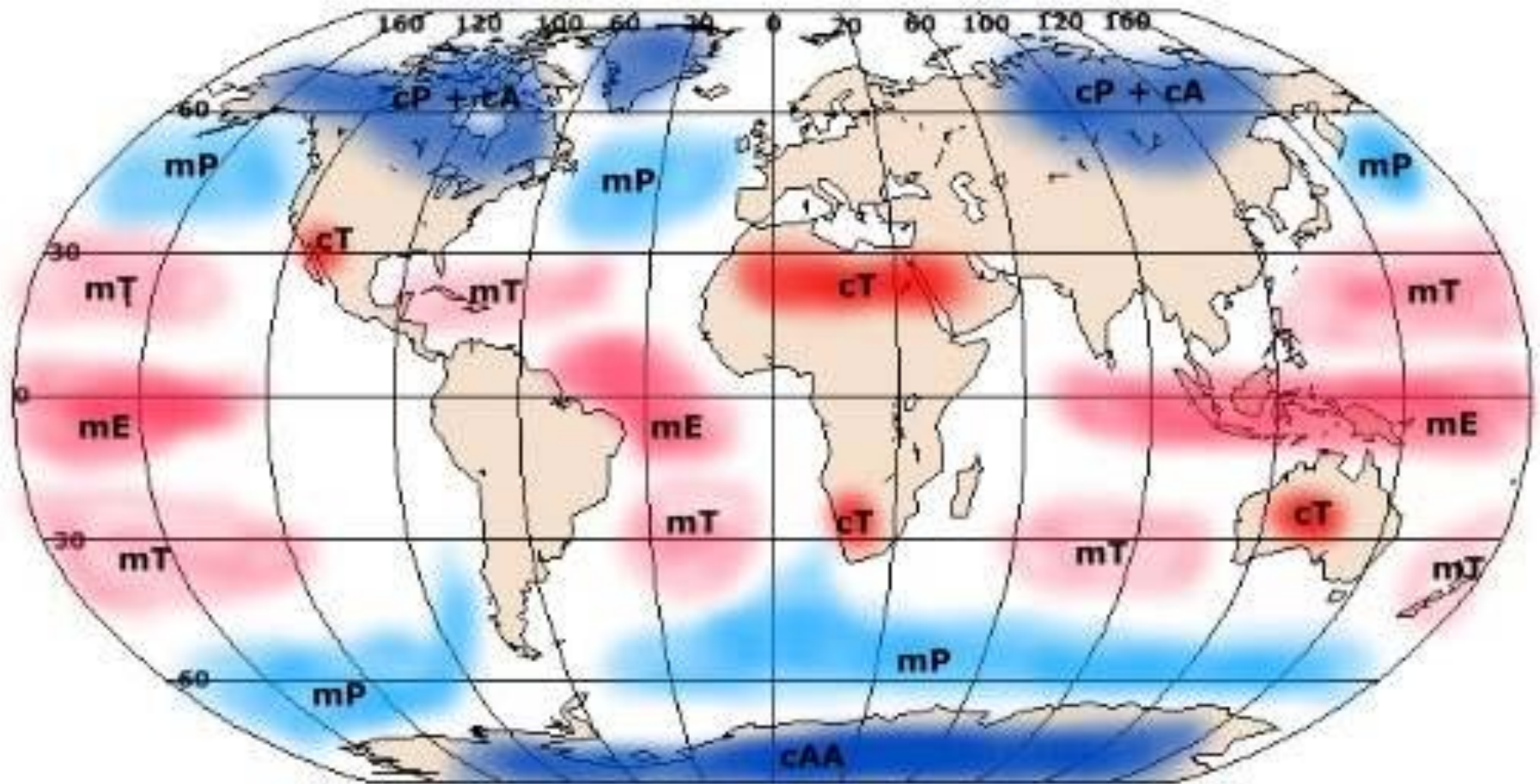


Arctic and Continental Arctic (A / cA)

- Extremely **low temperatures and very little moisture**
- Usually originate north of the Arctic Circle, where days of 24 hour darkness allow the air to cool to sometimes record-breaking **low temperatures**
- Form **over ice- and snow-covered regions of Siberia, the Arctic Basin, Greenland, and northern North America.**
- These air masses often **plunge south across Europe during winter**, but very rarely form during the summer because the sun warms the Arctic.



Global Air Masses (Popular ones):



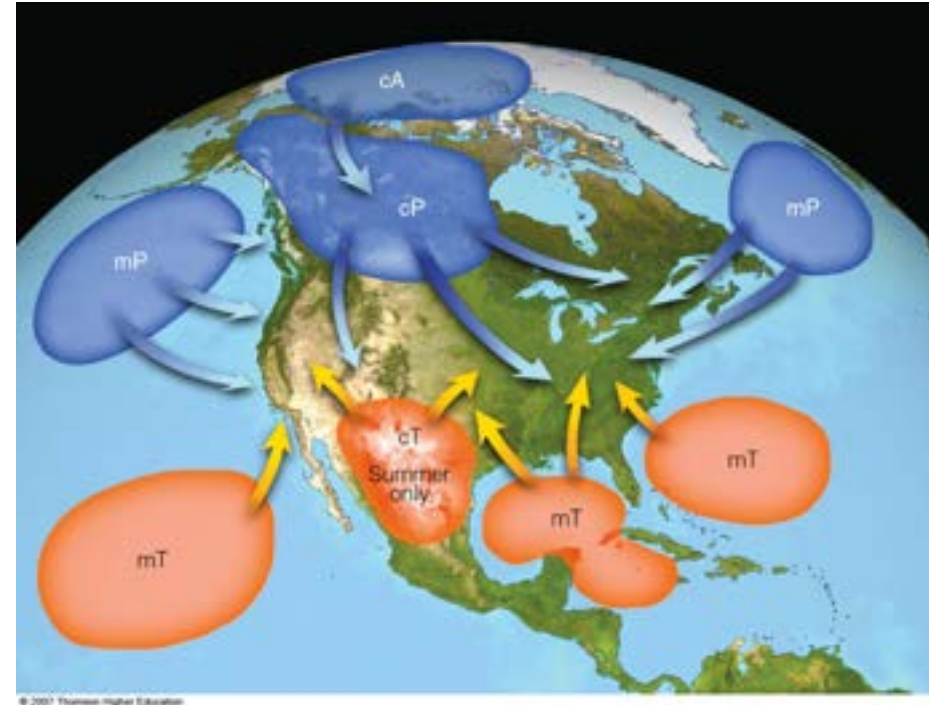
Source Regions & Classification (Regional case)

cP: cold, stable air originating from high pressures in N. Canada

mP: cool, moist air, usually unstable originating from Aleutian low; region of cyclone formation; heavy precipitation in coastal ranges

mT: warm moist unstable air originating from the Gulf of Mexico; good for producing thunderstorms

cT: hot dry air from Mexico; impacts southern AZ and CA



Air Mass Characteristics

Air Mass Type	Weather Map Symbol	Source Region	Winter Characteristics	Summer Characteristics
Arctic	A	Siberia, Arctic Basin	Bitter cold, dry	Cold, dry
Continental polar	cP	Interiors of Canada and Alaska	Very cold, dry	Cold, dry
Continental tropical	cT	Southwest U.S., Mexico	Warm, dry	Hot, dry
Maritime polar	mP	North Pacific	Mild, humid	Mild, humid
		North Atlantic	Cold, humid	Cold, humid
Maritime tropical	mT	Gulf of Mexico, Caribbean Sea, tropical and subtropical Atlantic and Pacific Ocean	Warm, humid	Hot, humid

How Air Masses move ?

Air masses do not remain in their source region indefinitely but move from place to place changing their temperature, humidity, and/or stability.

- ❖ **Global circulation (winds)** usually move these masses of air from one part of the globe to another.
- ❖ **The major wind belts usually push air masses from the west to the east (vis-a-versa) across the region** depends on altitude.
- ❖ **Jet streams** also push air masses in the same west-to-east direction.
- ❖ As these air masses move, they will eventually collide with one another-resulted into transitional zones ??.
- ❖ Storms and other types of weather develop along the line where two different air masses collide.

The modification in air mass occurs primarily by

- i) exchanging heat or moisture, or both, with the surface over which the air mass travels
- ii) radiative cooling, and
- iii) processes associated with large-scale vertical winds.

Air Mass Modification Processes

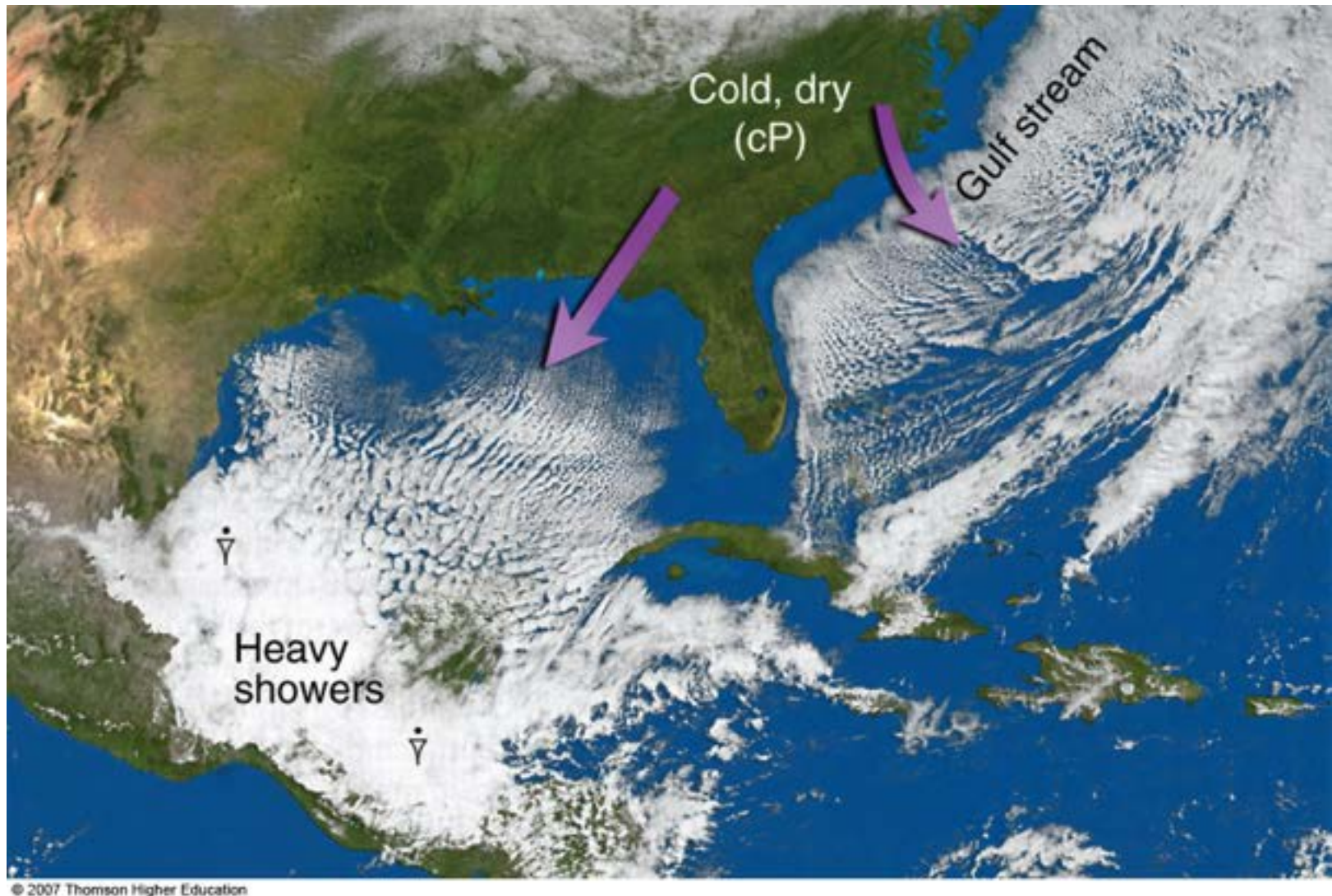
Thermodynamic

- **Surface heating/cooling**
 - Change of temperature of surface, or advection over different surface
- **Addition of moisture**
 - Surface evaporation
 - Evaporation of precipitation falling from higher level
- **Loss of moisture**
 - Condensation, precipitation
- **Radiative heating/cooling**
 - slow compared to surface heat exchange (up to 2 weeks)

Dynamic

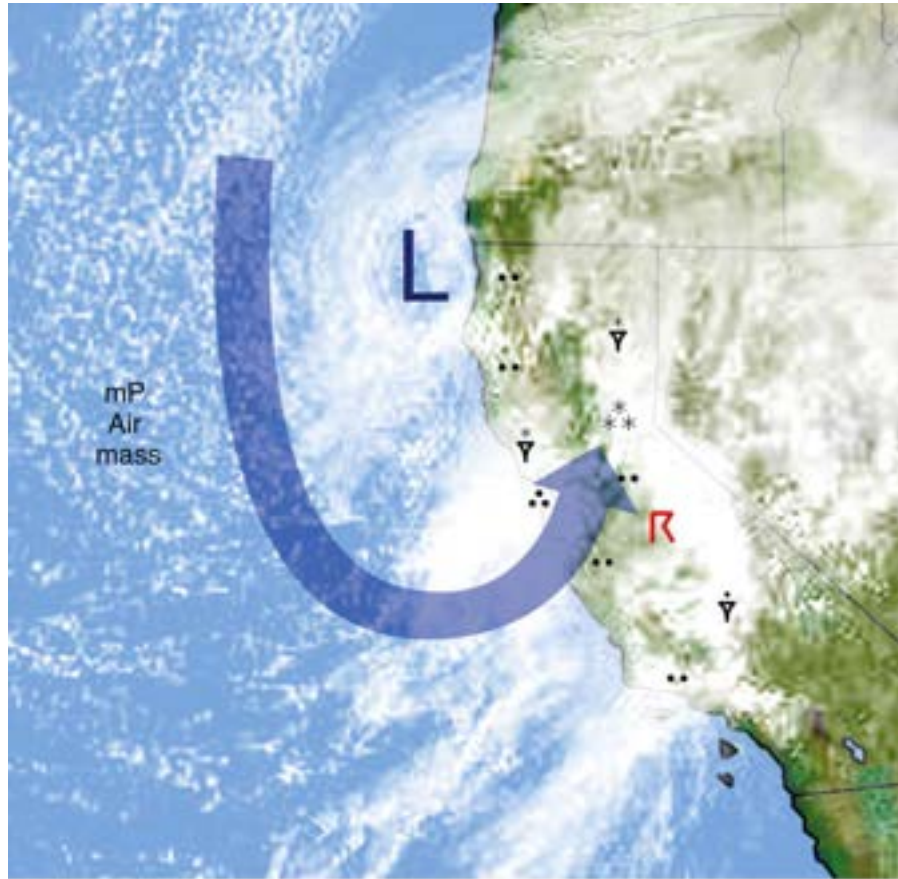
- **Turbulent mixing**
 - Increases uniformity of air mass. Very efficient close to surface.
- **Large-scale lifting/descent**
 - Causes adiabatic changes of temperature
 - May result in formation or evaporation of clouds

Modification of cP Air



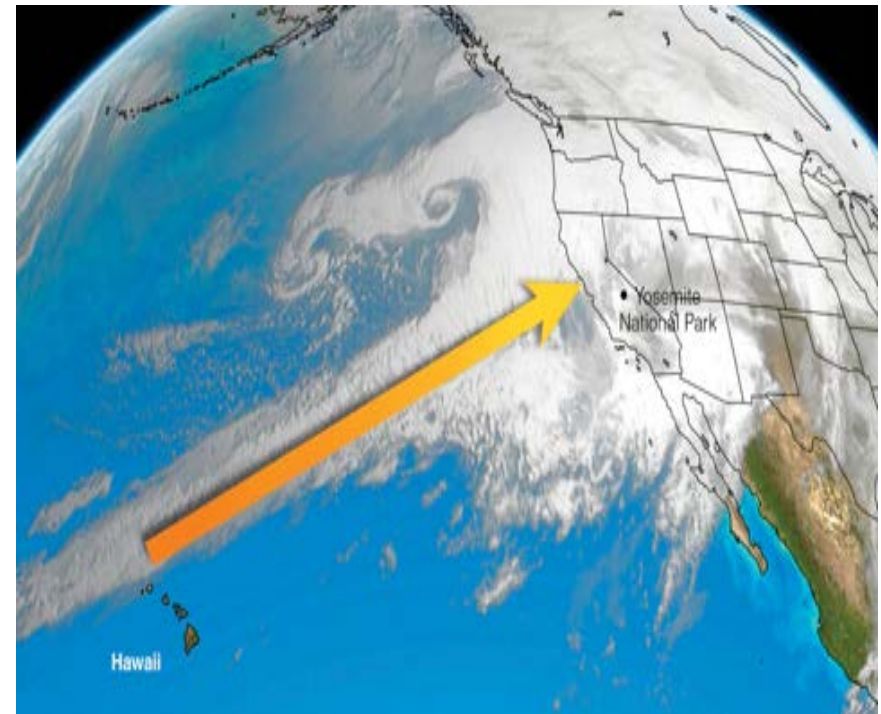
As the cP air mass moves over the warmer Gulf of Mexico and Gulf Stream, **ocean surface warmed the cooler air**. The lower atmosphere becomes **unstable and forms extensive rows of cumulus cloud streets**.

mP Air Masses

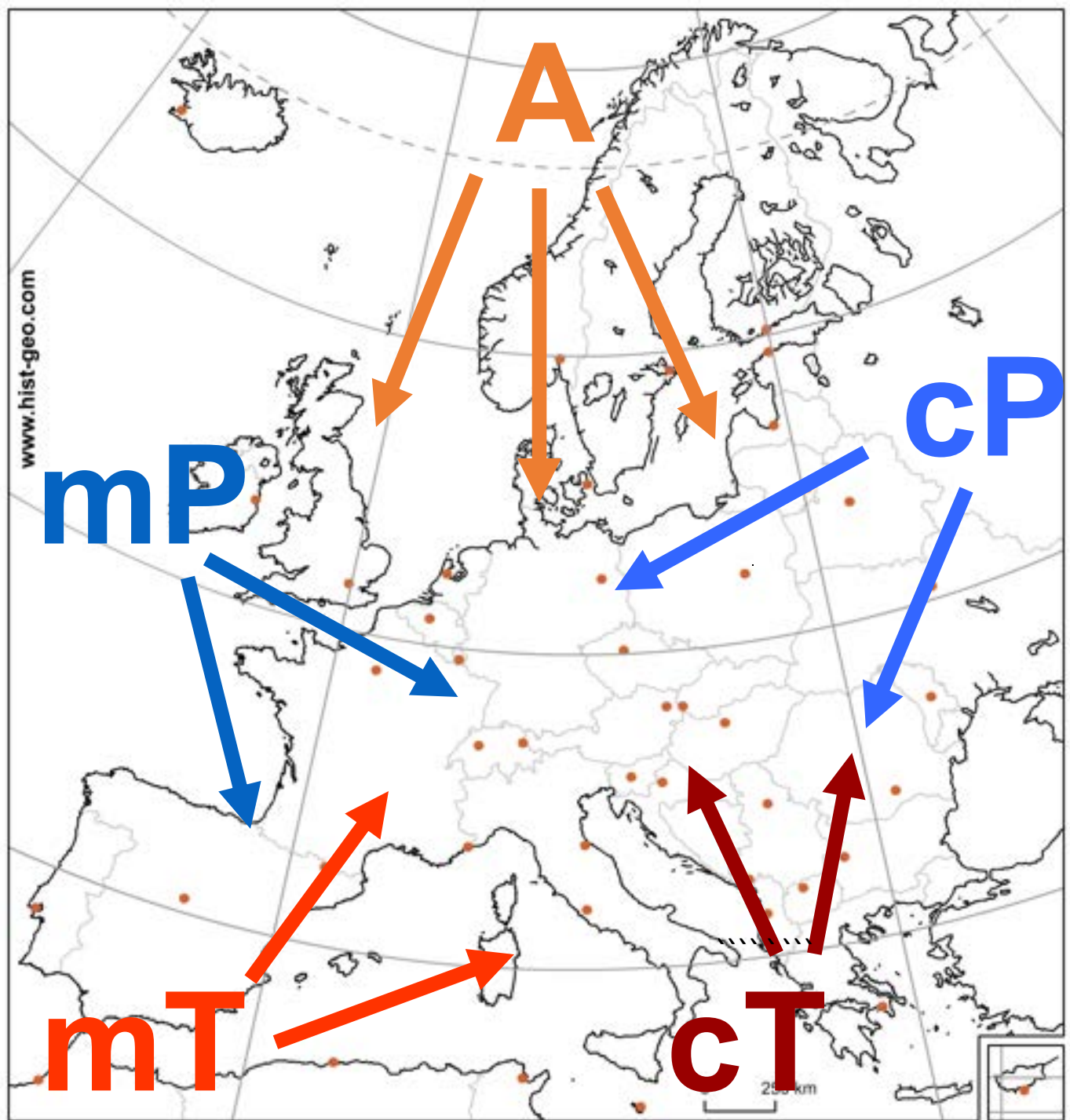


© 2007 Thomson Higher Education

mT Air Masses



**Air
masses
that affect
weather
in Europe**



How do you read weather symbols?

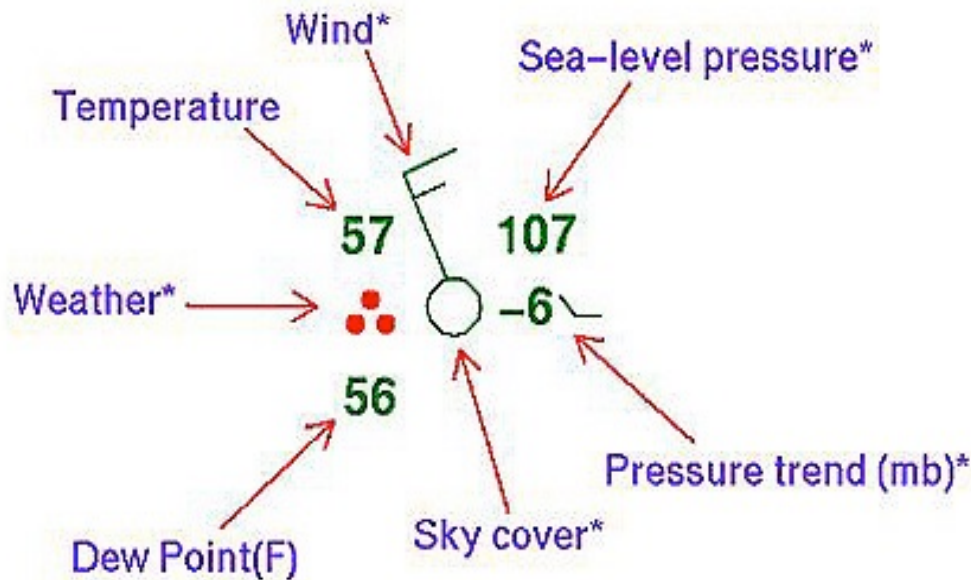
It represents data recorded from a weather station.

The circle is cloud cover.

The “tail” is the wind direction.

The feathers are the wind speed in knots. (1 knot = 1.15 miles)

Wind Speed	
☉	calm
—	1-2
—	5
—	10
— /	15
— /	20
— /	25
— /	30
— / /	50
— / /	65 knots



What is the percent cloud cover at the station?

0%

What is the wind speed in knots?

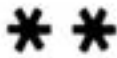
15 knots

What direction is the wind coming FROM?

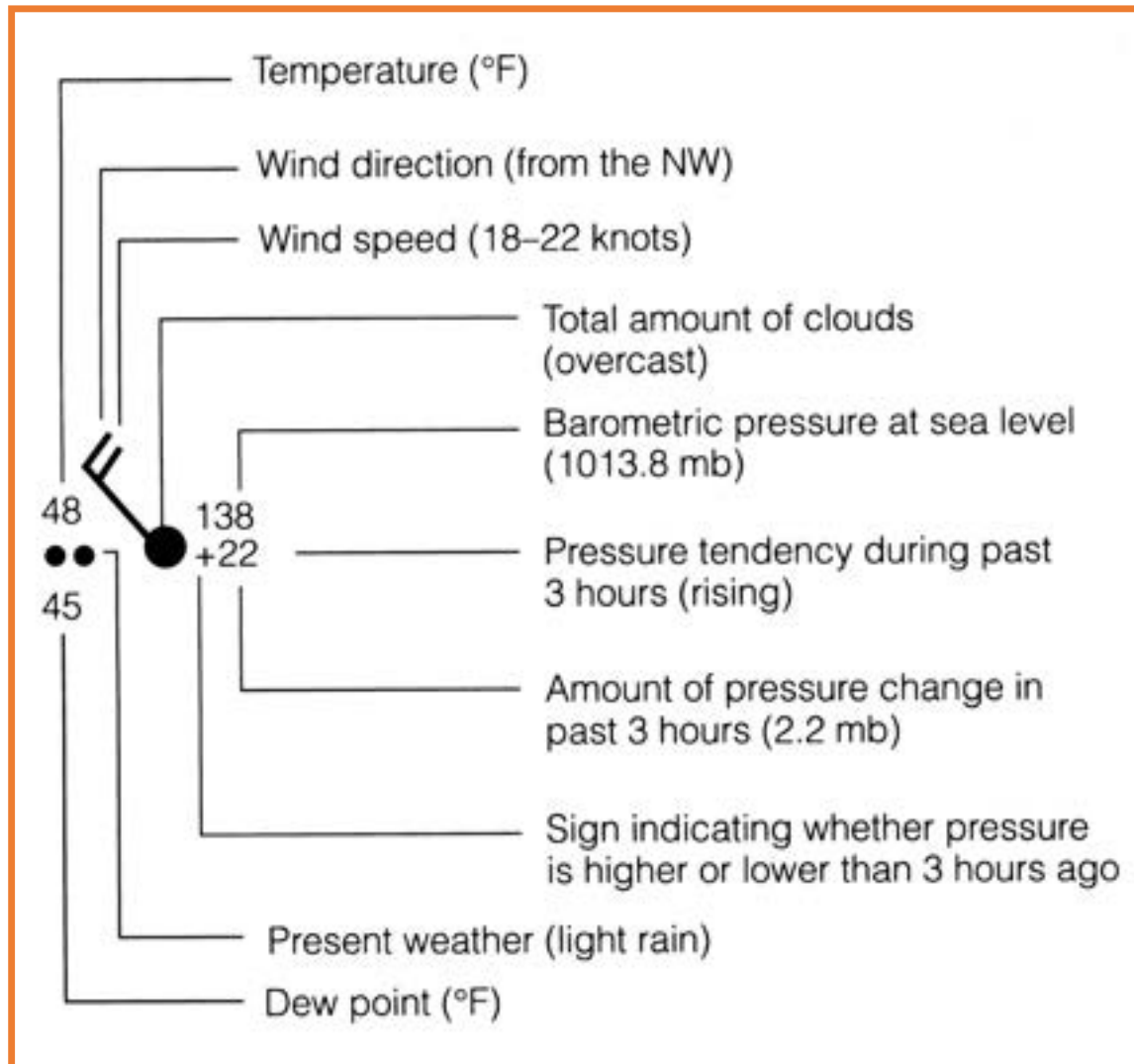
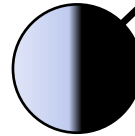
NW

What direction is it going to?

SE

	Light rain		Rain shower
	Moderate rain		Snow shower
	Heavy rain		Showers of hail
	Light snow		Drifting or blowing snow
	Moderate snow		Dust storm
	Heavy snow		Fog
	Light drizzle		Haze
	Ice pellets (sleet)		Smoke
	Freezing rain		Thunderstorm
	Freezing drizzle		Hurricane

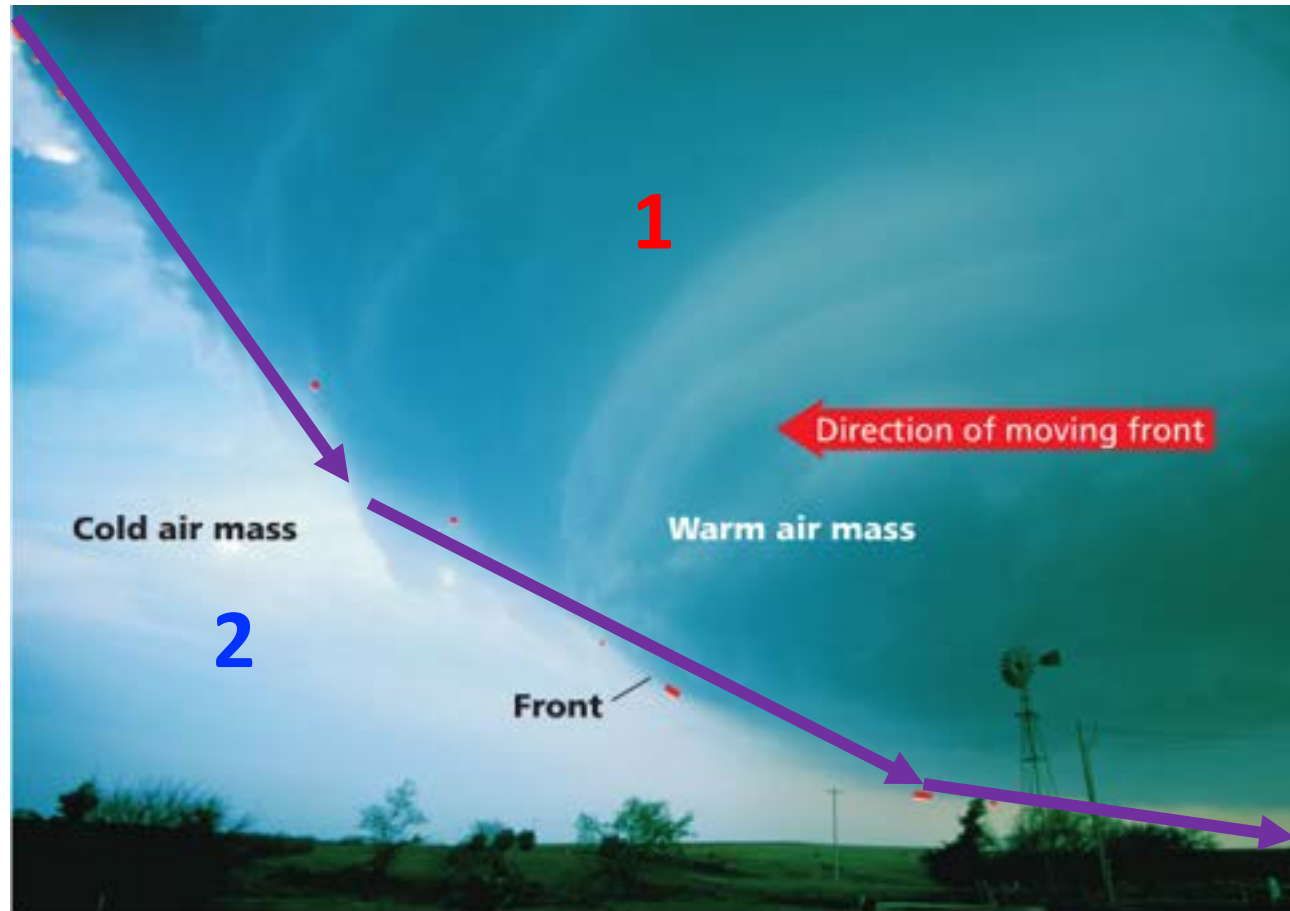
½ cloud cover
wind at 5 knots out of
the northeast.



	Clear
	1/8
	Scattered
	3/8
	4/8
	5/8
	Broken
	7/8
	Overcast
	Obscured
	Missing

Front

When two air masses that **DO NOT** have the same properties meet, they do not mix. Instead, a boundary forms that is called a front.



☞ A transition zone between two air masses of different physical properties (e.g. temperature and/or humidity).

The formation of the front is called- Frontogenesis

- ❑ involves a convergence of two distinct air masses.

The Decay/dissipation of the front is called –Frontolysis

- ❑ involves overriding of one of the air mass by another.



- In northern hemisphere Frontogenesis (convergence of air masses) happens in **anti-clockwise direction** and in southern hemisphere, **clockwise direction**.
- Mid-latitude cyclones or temperate cyclones or extra-tropical cyclones occur due to **frontogenesis**.

Characteristics of Fronts

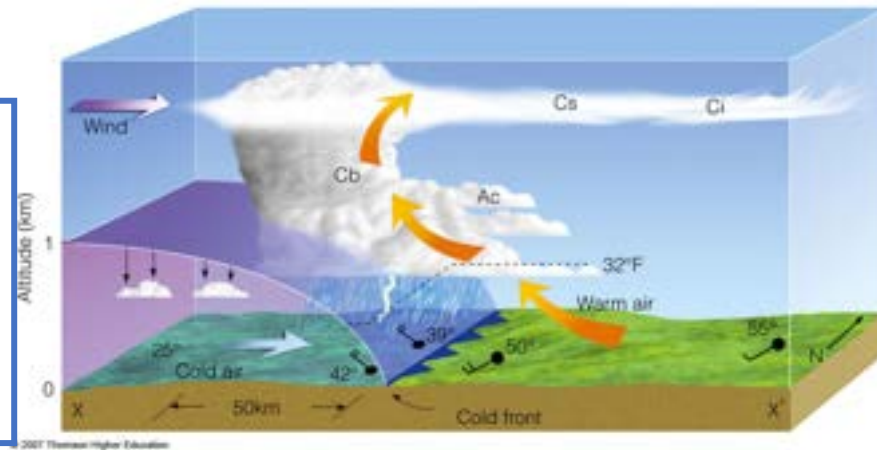
- Fronts are more **narrow** than air masses
- Fronts are the basic building blocks of weather systems.
- Typical occurrence of the frontal activity in midlatitudes weather regimes (region: 30°-65° N and S). They are **uncommon** (unusual) in tropical and polar regions.
- Fronts are often associated with **cloudiness and some form of precipitation**

Types of Fronts

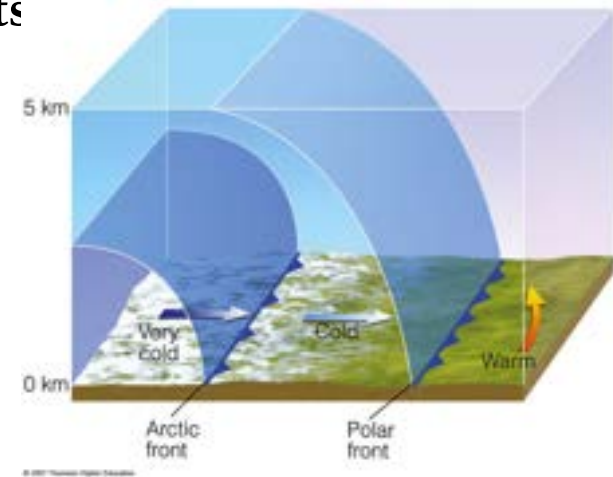
1. Cold front
2. Warm front
3. Occluded front
4. Stationary front

Cold front:-

- ☛ A transition zone where a cold air mass advances and replaces a warm air mass.
- ☛ Remember that warm air will rise up over the cold air because the warm air is **less dense**.

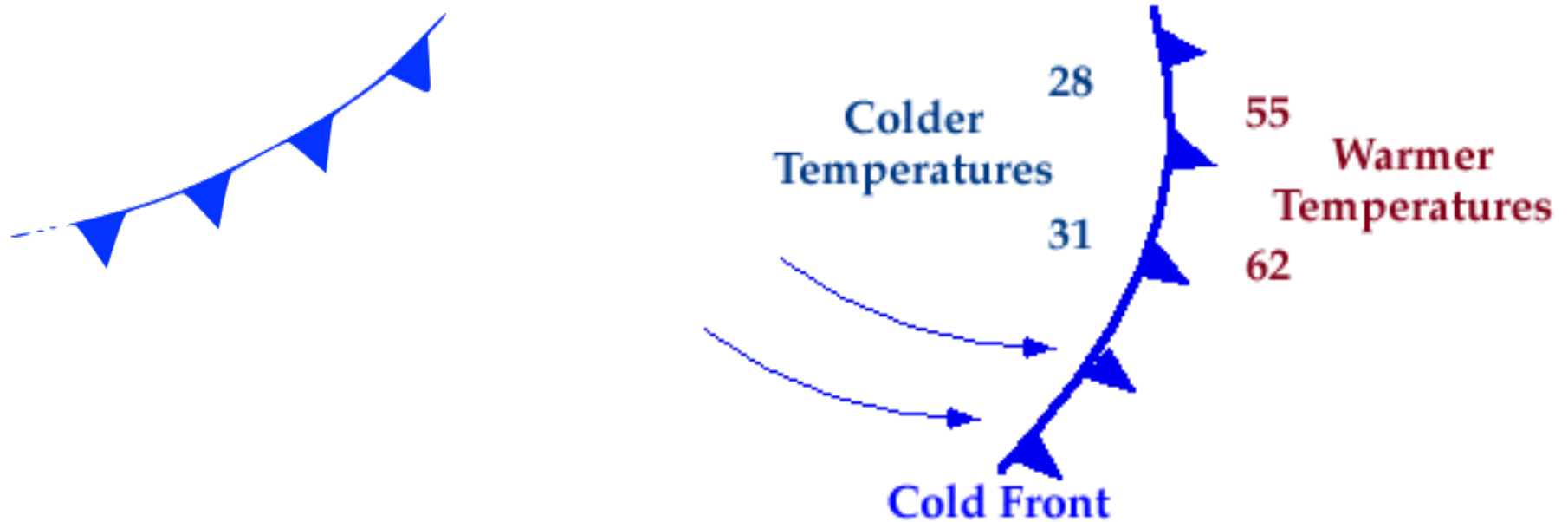


- ☛ Violent weather is associated with a cold front and then fair, cool weather follows.
- ☛ Thunderstorms and tornadoes form with cold fronts
- ☛ Precipitation will usually occur behind the front
- ☛ Vertical cross section: slope of 1:50 (sharp slope)
- ☛ short duration (hours)
- ☛ Temperatures drop anywhere from 5° to 15°.
- ☛ Wind speed is around 20 mph
- ☛ Winds become gusty and erratic.



Weather Map - Cold front:-

A cold front on a weather map is indicated by a blue line with triangles pointed in the direction that the front is moving towards



Cold front - Animation

https://mesoscale.agron.iastate.edu/agron206/animations/05_cnWfronts.html

Factors- Cold Front Weather- during the evolution

• **TABLE 11.2**

Typical Weather Conditions Associated with a Cold Front in the Northern Hemisphere

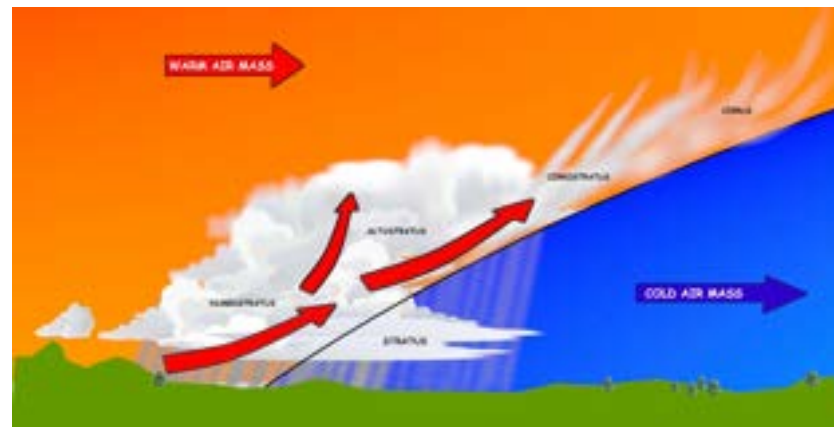
WEATHER ELEMENT	BEFORE PASSING	WHILE PASSING	AFTER PASSING
Winds	South or southwest	Gusty, shifting	West or northwest
Temperature	Warm	Sudden drop	Steadily dropping
Pressure	Falling steadily	Minimum, then sharp rise	Rising steadily
Clouds	Increasing Ci, Cs, then either Tcu* or Cb*	Tcu or Cb	Often Cu, Sc* when ground is warm
Precipitation	Short period of showers	Heavy showers of rain or snow, sometimes with hail, thunder, and lightning	Decreasing intensity of showers, then clearing
Visibility	Fair to poor in haze	Poor, followed by improving	Good, except in showers
Dew point	High; remains steady	Sharp drop	Lowering

*Tcu stands for towering cumulus, such as cumulus congestus; whereas Cb stands for cumulonimbus. Sc stands for stratocumulus.

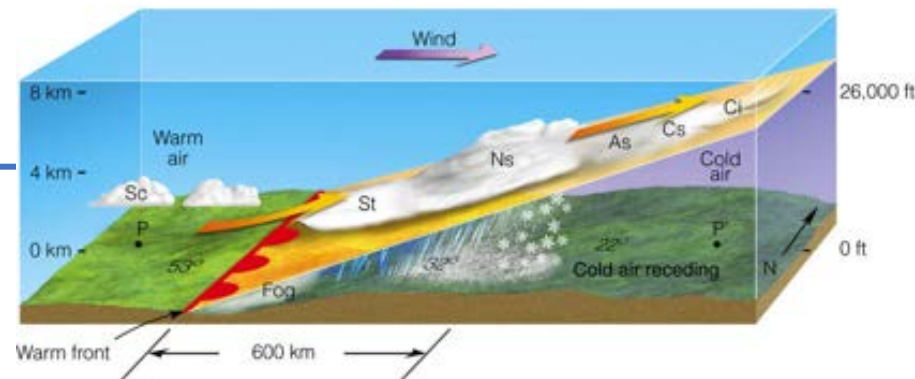
Most cold fronts move toward the south, southeast, or east.

Warm front:-

- ☞ A warm front occurs when a warm air mass meets and overrides a cold air mass. Here cold air retreats.
- ☞ The warm air gradually replaces the cold air.

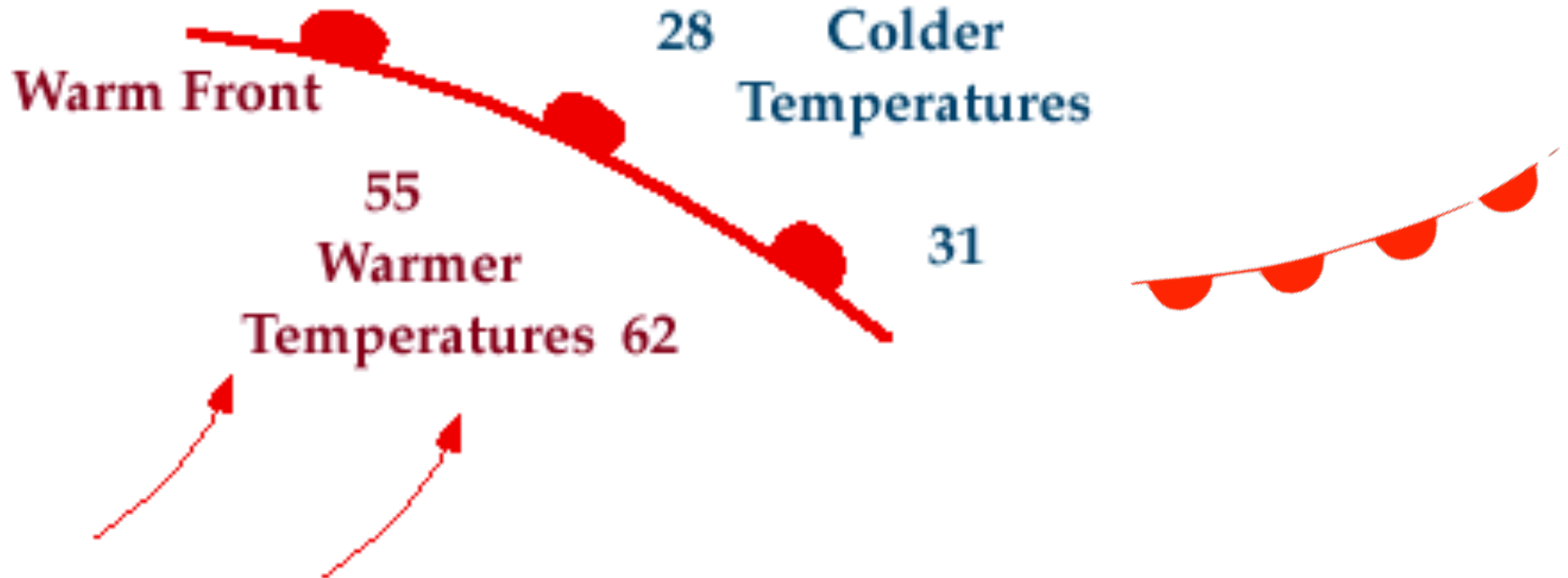


- ☞ This lifting is generally much more smooth and slow than a cold front.
- ☞ Warm fronts generally bring mild and drizzly precipitation.
- ☞ Wind speeds are 10 mph
- ☞ leading edge is not steep (1:150)
- ☞ Slightly more Duration (days) than cold front (hours)
- ☞ Temperature and humidity differences not as large as those for cold fronts



Weather Map - Warm front:-

A cold front on a weather map is indicated by a red line with “semi-circles” pointed in the direction that the front is moving towards



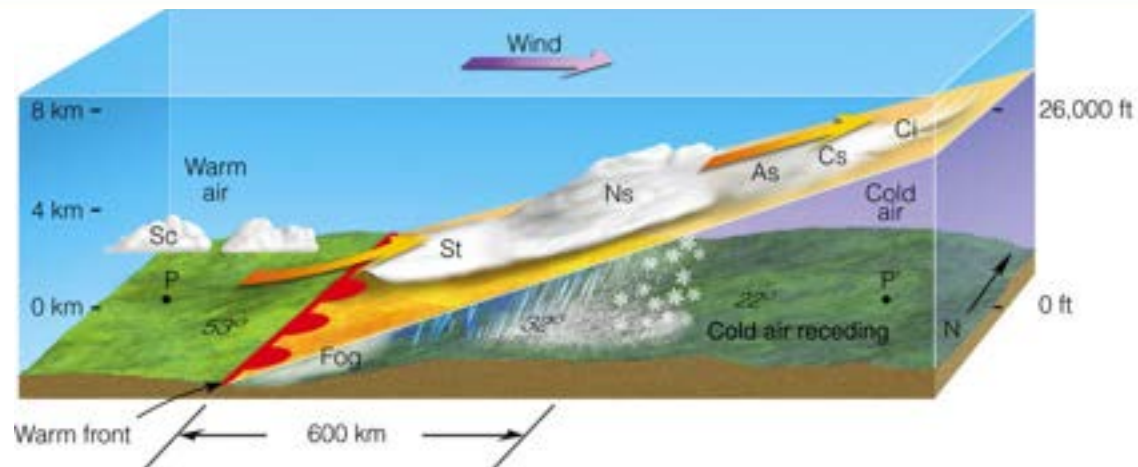
Warm front - Animation

https://mesoscale.agron.iastate.edu/agron206/animations/05_cnWfronts.html

Factors- Warm Front Weather- during the evolution

■ **TABLE 8.3** Typical Weather Conditions Associated with a Warm Front in the Northern Hemisphere

WEATHER ELEMENT	BEFORE PASSING	WHILE PASSING	AFTER PASSING
Winds	South or southeast	Variable	South or southwest
Temperature	Cool to cold, slow warming	Steady rise	Warmer, then steady
Pressure	Usually falling	Leveling off	Slight rise, followed by fall
Clouds	In this order: Ci, Cs, As, Ns, St, and fog; occasionally Cb in summer	Stratus-type	Clearing with scattered Sc, especially in summer; occasionally Cb in summer
Precipitation	Light-to-moderate rain, snow, sleet, or drizzle; showers in summer	Drizzle or none	Usually none; sometimes light rain or showers
Visibility	Poor	Poor, but improving	Fair in haze
Dew point	Steady rise	Steady	Rise, then steady



Occluded front:-

☞ A warm air mass sometimes is caught between two cooler air masses OR a cold front catches up to a slower-moving warm front. This forms an occluded front.

☞ A complex frontal system

When the air behind the front is colder than the air ahead of it, the front is called a **cold occlusion**.

When the air behind the front is milder than the air ahead of it, the front is called a **warm occlusion**.

☞ This produces cool temperatures and large amounts of precipitation

☞ Temperature fluctuations are small and winds are gentle.

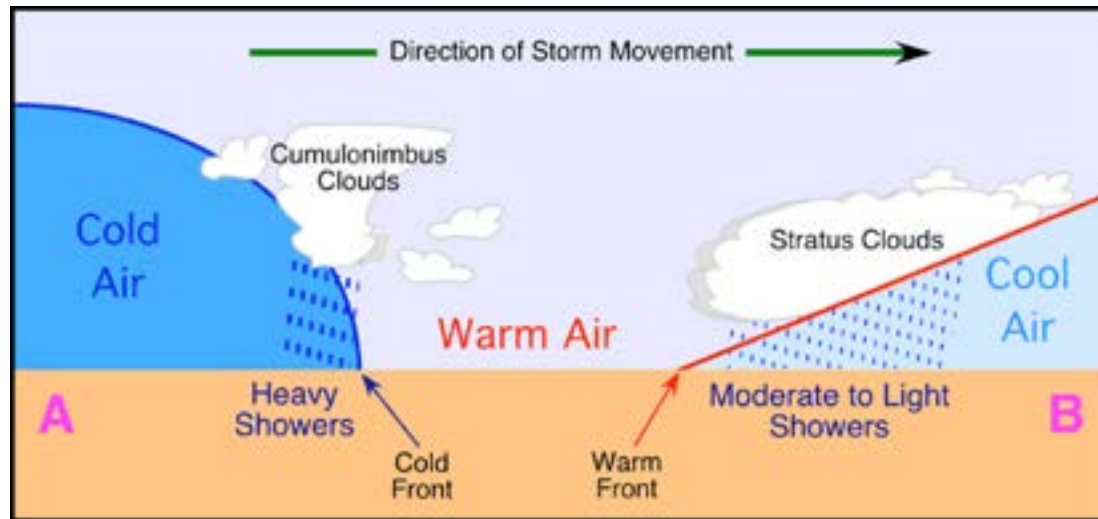
☞ May indicate the end of a storm cycle

☞ Produces less extreme weather than a cold front or a warm front.

☞ Occluded fronts have characteristics of both warm and cold fronts.

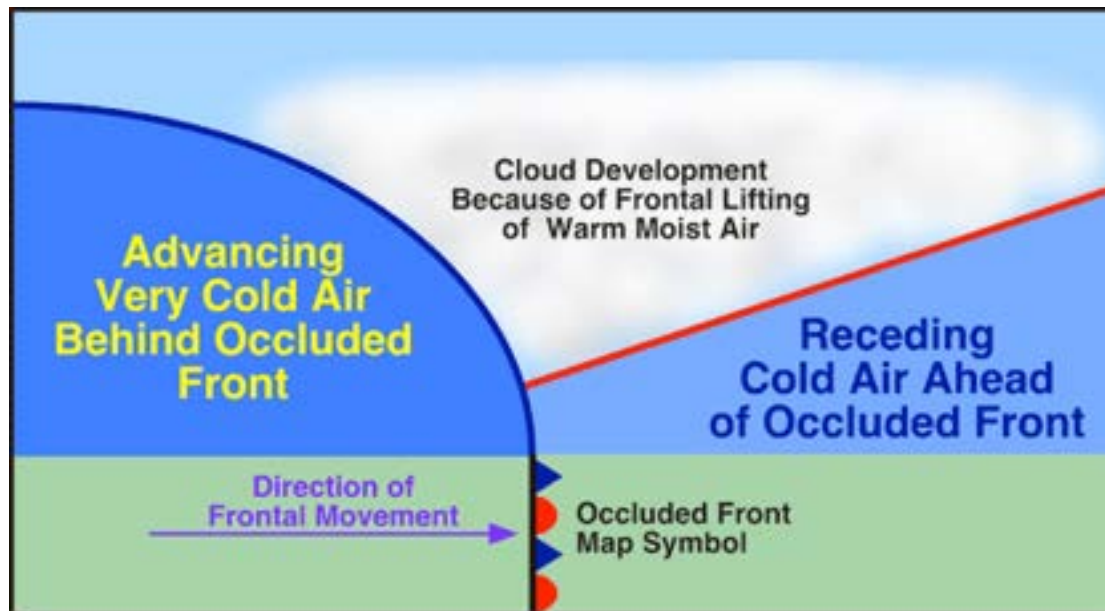
...After occlusion, the front would die out

Step 1



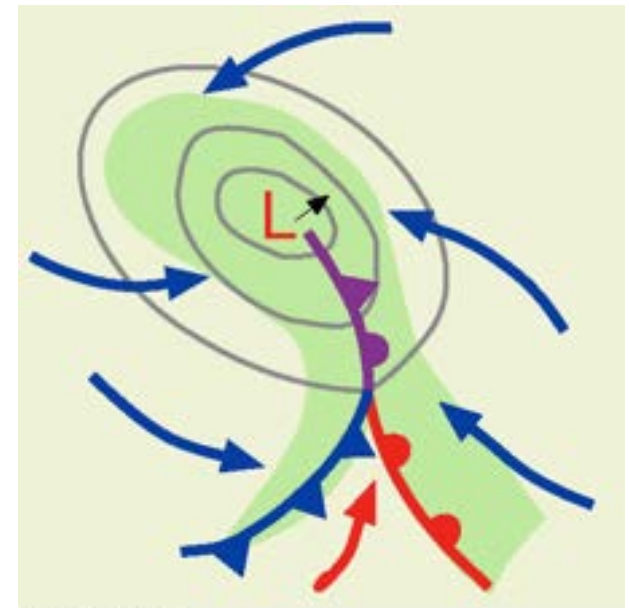
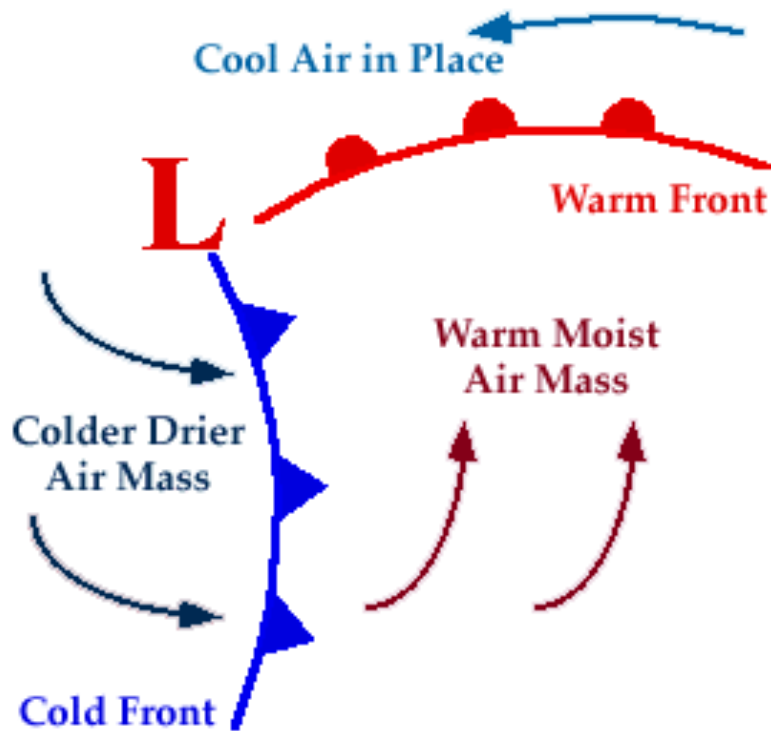
- lifts the warm air (behind the warm front) off the ground, creating an occluded front, where the two fronts are joined.

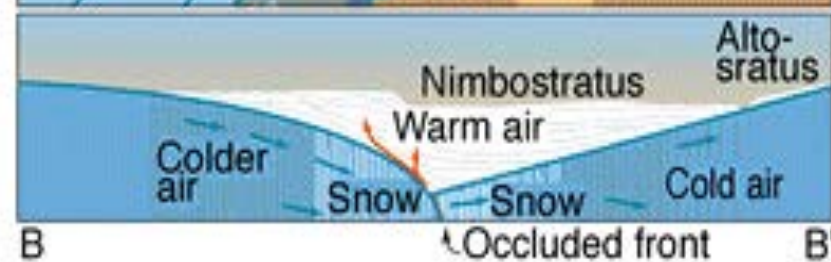
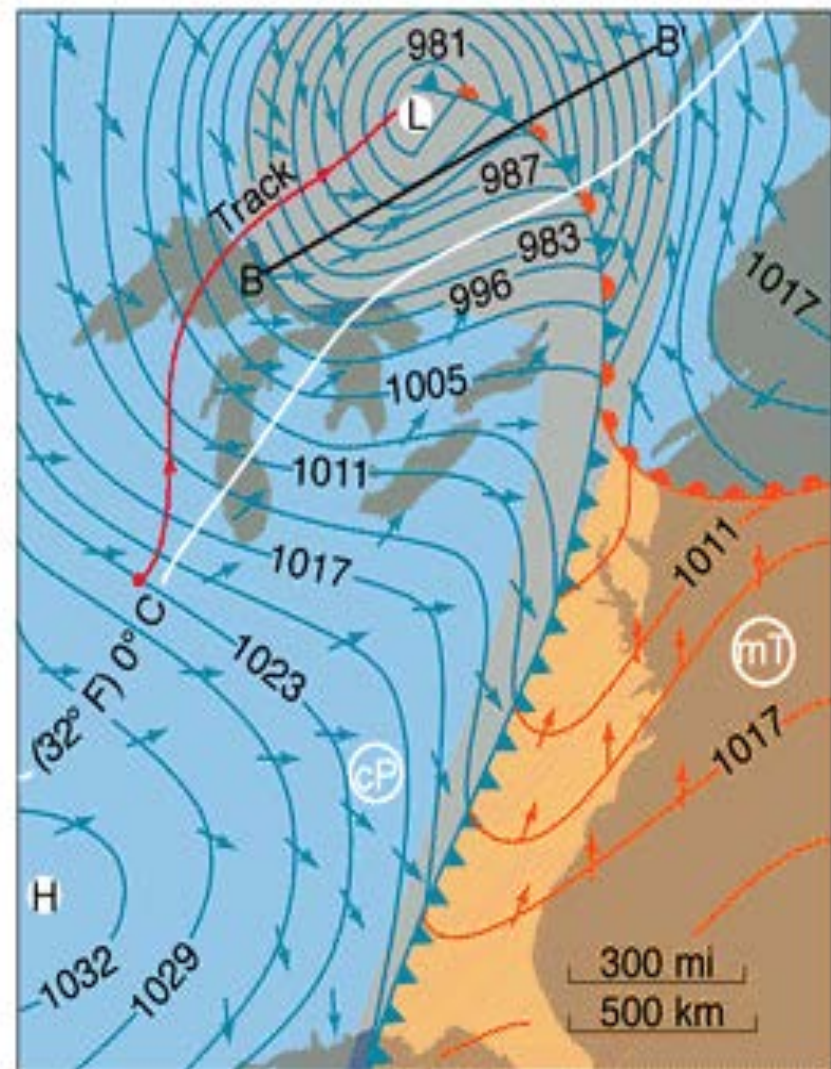
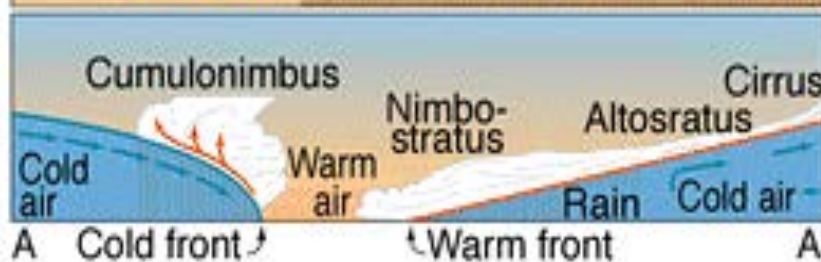
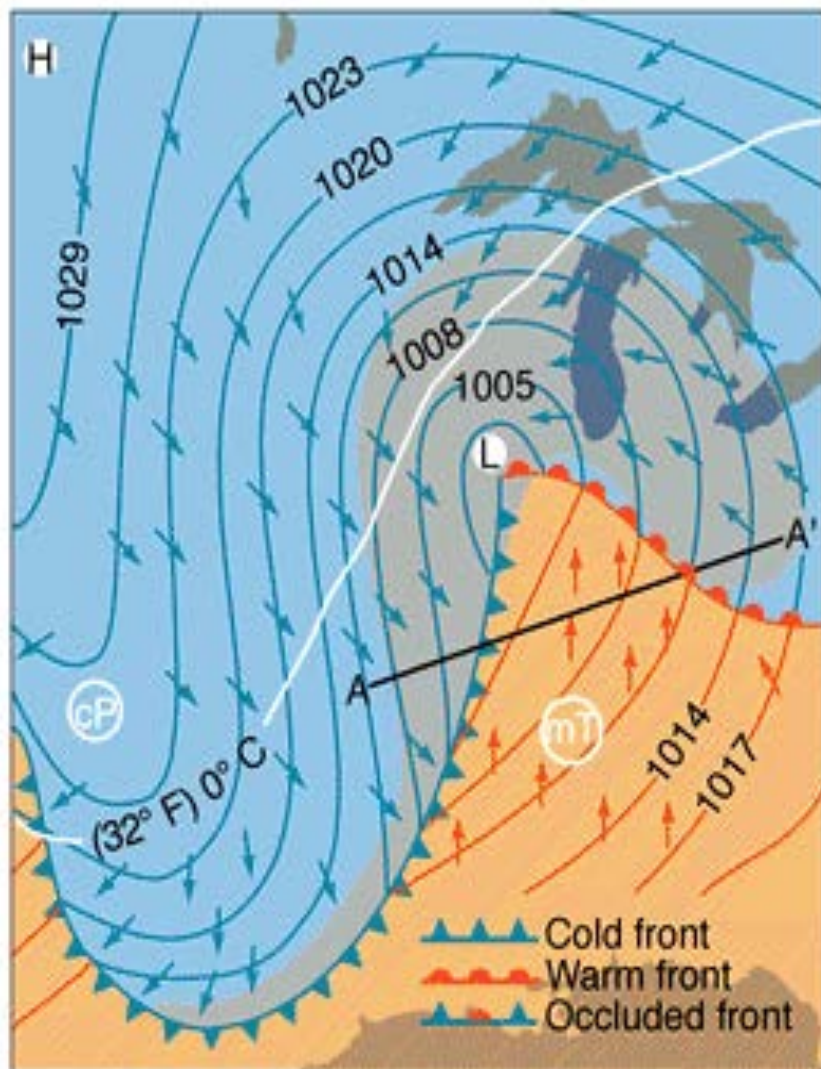
Step 2



Weather Map - Occluded front:-

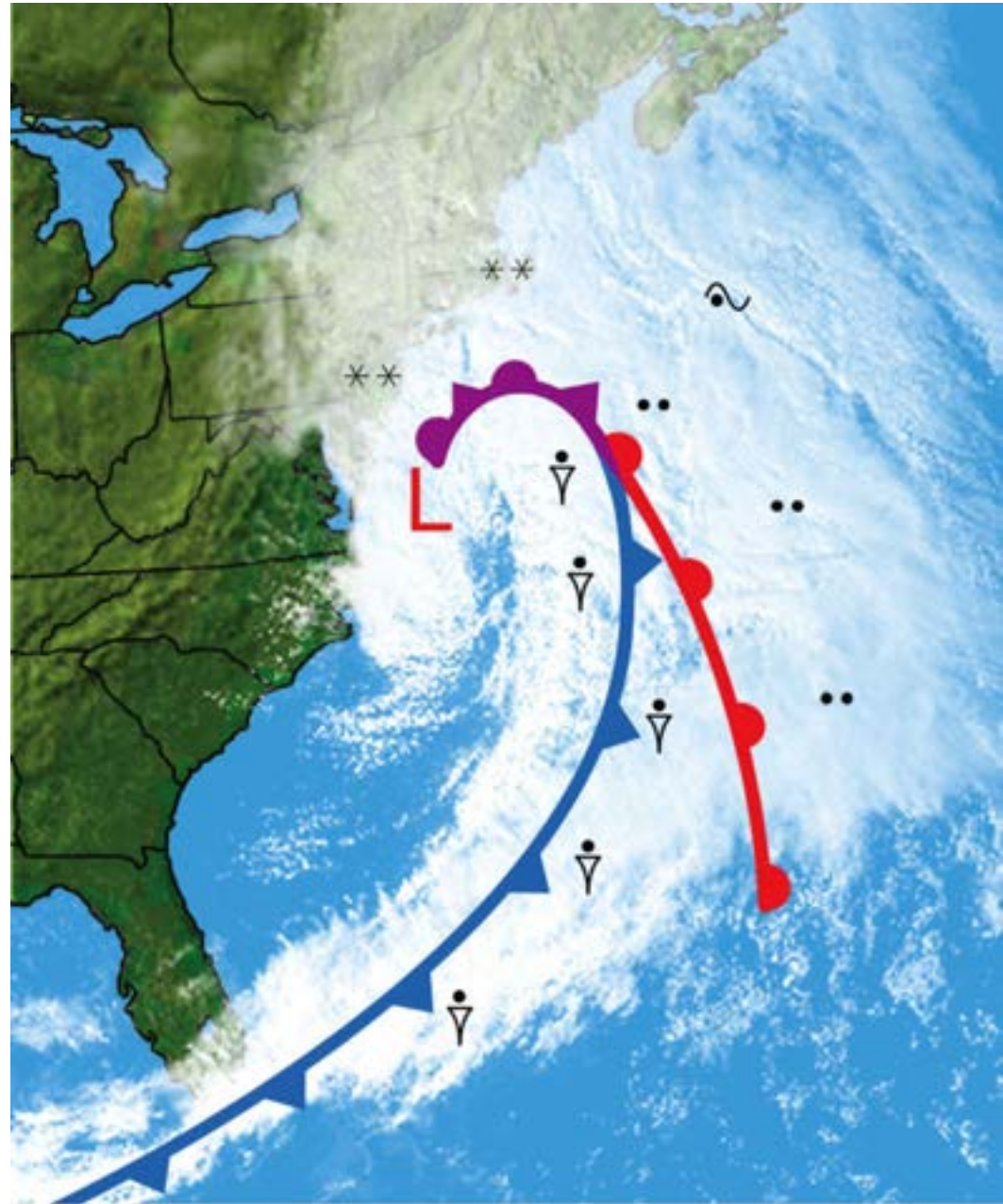
This front on a weather map is indicated by **as purple lines of alternating triangles and semicircles** pointed in the direction that the front is moving towards





Mid-Latitude Cyclone

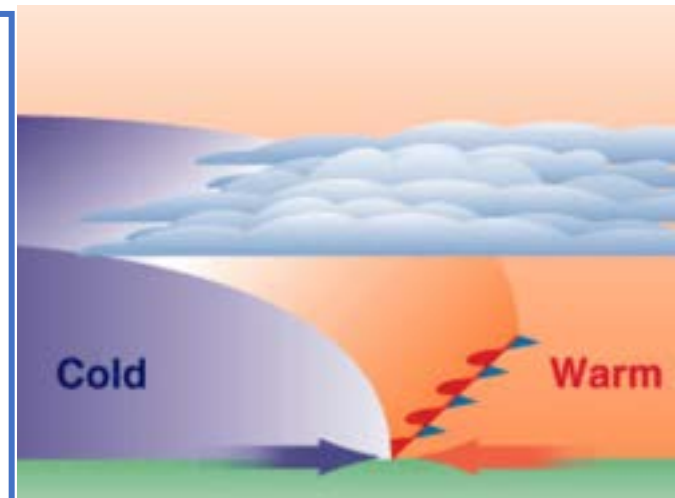
- Occluded fronts are common along mid-latitude cyclones, or deep low pressures centers about which the cold and warm fronts pivot.
- Mid-latitude cyclones are also known as extratropical cyclone.
- These storms appear frequently in the mid-latitudes.
- These storms are formed as a result of colliding air masses (polar front theory).

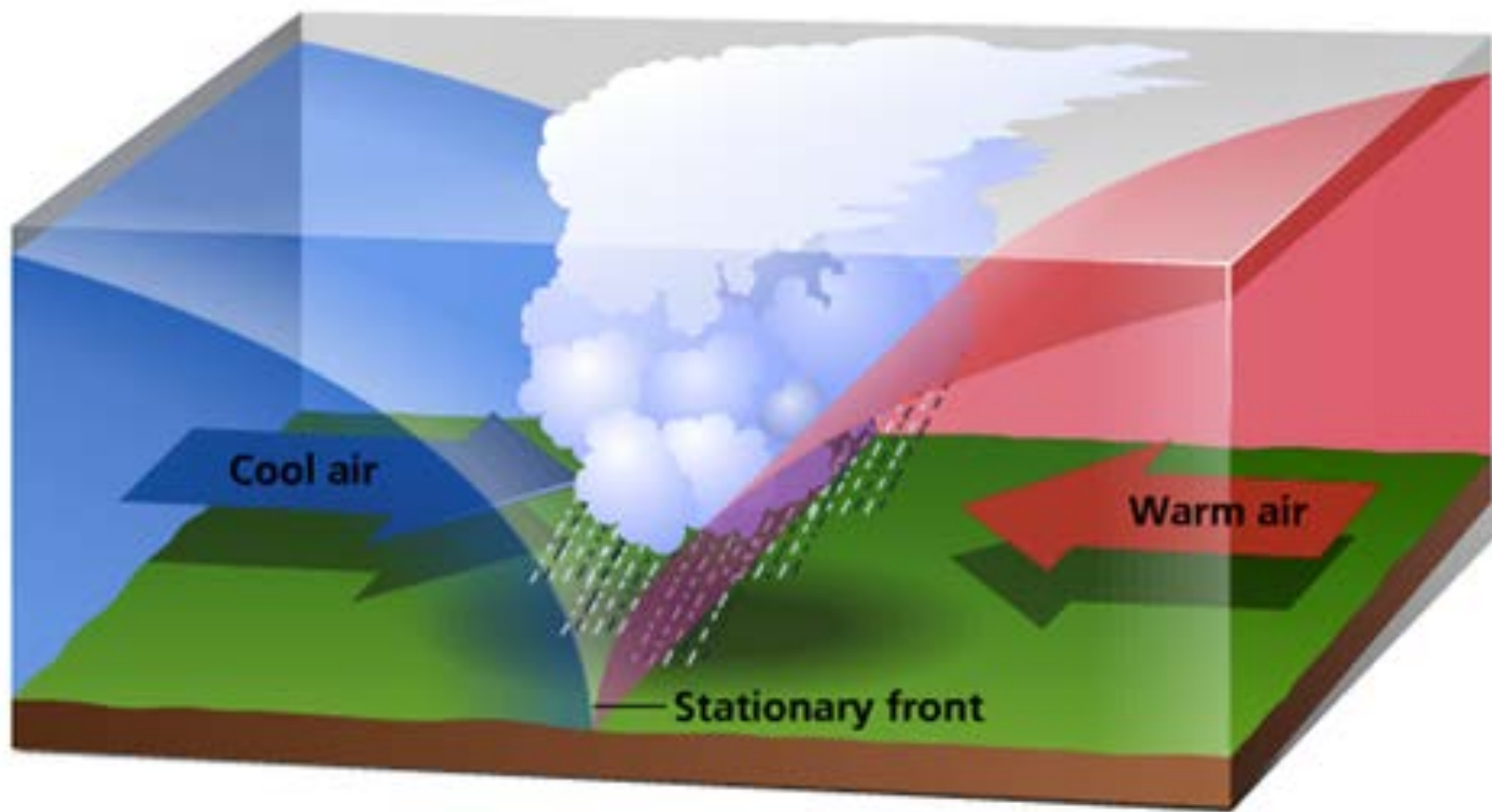


Stationary front:-

- ☞ Sometimes cold and warm air masses meet, but neither air mass moves over or under the other, they may even travel parallel to the frontal boundary.
- ☞ A front that is nearly stationary with winds blowing parallel and from the opposite directions on each side of the front at the surface. This is known as a stationary front.

- ☞ Temperatures remain stagnant and winds are gentle to nil.
- ☞ Weather produced is similar to a warm front.
- ☞ A stationary front can bring many days of clouds and precipitation.
- ☞ These type of conditions can last for days, producing nothing but Altocumulus clouds.

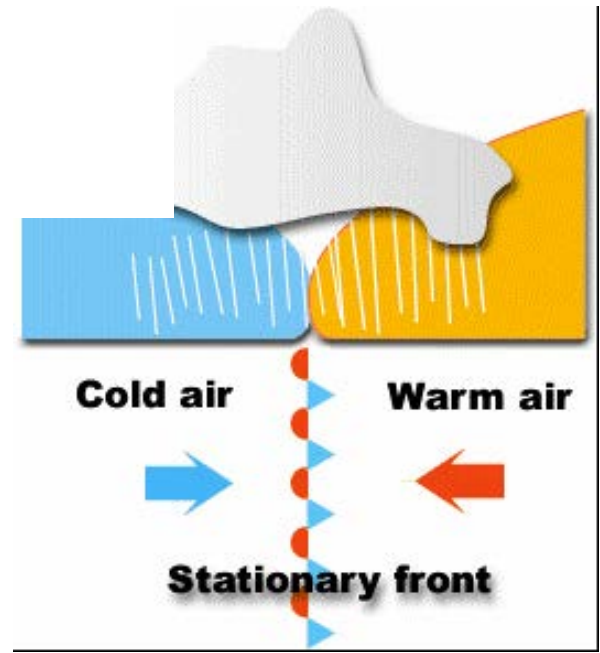




Weather Map - Stationary front:-

This front on a weather map is indicated by **inverting** blue/red triangles/semicircles.

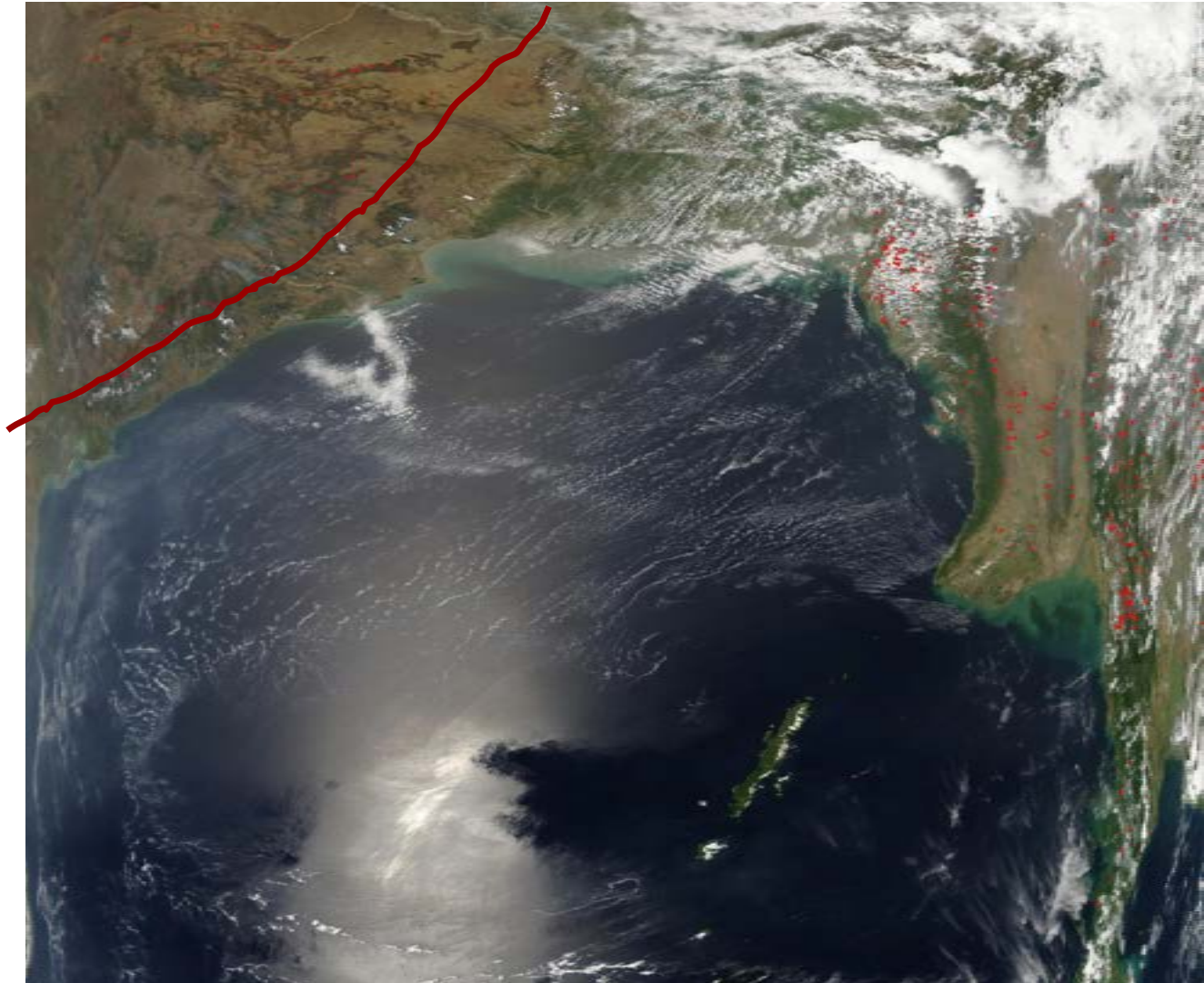
Stationary Front



Dryline:-

A dryline is a boundary that separates a moist air mass from a dry air mass

Dryline over eastern India



primary difference in moisture